

DISCRETE TRIAL TRAINING

CHARACTERISTICS OVERVIEW CHART

Verbal Skills	Grade Levels	Cognitive Level	Areas Addressed
☒ Nonverbal	☒ PK	☒ Classic	☒ (Pre)Academic/Cognitive/Academic
☒ Mixed	☒ Elementary	☒ High Functioning	☒ Adaptive Behavior/ Daily Living
☒ Verbal	☒ Middle/High		☒ Behavior ☒ Communication/Speech ☒ Social/Emotional

BRIEF INTRODUCTION

Discrete trial training (DTT) is an applied behavior analysis strategy that focuses on skill acquisition by manipulating the sequence of antecedents and consequences. The main components of discrete trial training include instruction, prompting, response, consequence, and inter-trial interval.

DESCRIPTION

Discrete trial training is a method of teaching new skills, consisting of a series of distinct repeated lessons or trials taught one-to-one. Lovaas (1981) defined a trial as a “single teaching unit.” A particular trial typically consists of four parts: (a) the discriminative stimulus (S^D) (the instruction), (b) the response (R), (c) the consequence or the reinforcing stimulus (S^R), and (d) the inter-trial interval (ITI) (presentation wait time). In addition, an optional prompt (S^P) may be used to help the child respond correctly. The parts of the discrete trial are often represented symbolically in an order like the following:



Essentially, first the teacher's instruction is given (S^D). A prompt, cue, or model from the teacher may be provided to help the child respond correctly (S^P). Then the child responds to the instruction, either with help or without (R). The child's response is evaluated as correct, incorrect, or no response. Last, a consequence based on the child's response relative to a predetermined criterion follows (S^R).

Afterward the teacher pauses a little while before continuing to let the child know that they have completed one set and have moved on to the next (I^{TH}).

A significant amount of research supports the use of discrete trial training with individuals with autism (AU) in a variety of settings. Discrete trial training can help to compensate for challenges faced by this population and is beneficial for them for the following reasons:

- In discrete trial training, tasks are broken down into short and simple trials that accommodate the needs of individuals with short attention spans.
- Discrete trial training attempts to build motivation by rewarding performance of desired behavior and completion of tasks with tangible or external reinforcement.
- Stimuli presented in discrete trial training are clear and relatively consistent. The child is given rewards only for behaviors in response to those stimuli.
- Discrete trial training teaches skills and behaviors explicitly (cause-effect learning).
- The instructions given in discrete trial training are simple, concrete, and clearly provide only the most salient information.
- Discrete trial training can be designed to teach perspective taking and social cognition skills explicitly.

Discrete trial training must be individually and carefully applied since each child with autism exhibits unique combinations of deficits and excesses in various areas (Lovaas, 1987; Smith, 2001). Furthermore, caution must be exercised not to adopt this strategy to the exclusion of all other interventions. In addition, if discrete trial training is used as an instructional strategy, generalization of the skill must be considered.

STEPS

1. *Instruction.* A teacher gives succinct and clear instruction to a student. The use of few and consistent words is preferred.
2. *Prompting.* Based on the student's learning level and the complexity of the task, the teacher gives appropriate prompts to trigger the desired response.
3. *Response.* The student responds to the teaching and prompting. The responding behavior must be measurable.
4. *Consequence.* The teacher gives immediate feedback and reinforcement after student's response to enhance skill acquisition.
5. *Inter-trial interval.* A clear wait time is inserted after a trial completion before moving onto the next trial.

BRIEF EXAMPLE

To teach 4-year-old Jin to identify colors, Jin and her teacher sat down at a table in a quiet corner of the classroom. After getting Jin's attention on task, the teacher presented Jin's "penny board," which Jin was familiar with, and told her clearly that if she got five correct answers, she would be allowed to choose a desired activity.

Three cards of different colors were presented on the table. The teacher stated the color of each card while pointing to the card. After repeating this sequence, the teacher scrambled the three cards and put them on the table for Jin to practice. The teacher pointed to a card and said, "What color is it?" Waiting for couple of seconds, the teacher prompted Jin by exaggerating the first sound of "blue," for example. When Jin gave the right answer, she received a "penny" and verbal praise from the teacher.

The practice continued to follow this structure of steps until Jin mastered the skill. The teacher recorded data on every response and administered reinforcement for each correct answer. When finishing the trial, they counted the "pennies" Jin had earned. Because she had earned

five “pennies,” she earned five minutes of play with a toy she had selected from her reinforcement menu.

SUMMARY

Discrete trial training is aimed at manipulating the sequences of antecedents and consequences in a structured setting. Most often used in a one-to-one setting, discrete trial training contains five distinct steps for skills acquisition. Regular data collection provides detailed information to guide future instruction.

RESEARCH TABLE

Number of Studies	Ages (year)	Sample Size	Area(s) Addressed	Outcome
28*	3-13	346	Communication, behavior generalization, parent discrete trial training training, object matching, PECS, vocal imitation, motor, person/social functioning, cognitive functioning, stereotype behaviors, parent education, social-emotional functioning, out-of-seat behavior, motor imitation, daily living skills, identifying emotions, identifying objects, IQ, joint attention, spontaneous responses, social interaction, social quotients, listener skills, adaptive behavior, play responses, math and language arts content, following directions to choose.	+

*Note. Includes a review of literature by Eldevik et al (2009).

STUDIES CITED IN RESEARCH TABLE

- Adcock, A., & Cuvo, A. J. (2009). Enhancing learning for children with autism spectrum disorders in regular education by instructional modifications. *Research in Autism Spectrum Disorders*, 3, 319-328.
Discrete trial training accompanied by a preference assessment, prompts, and reinforcement was used to teach children with autism (ages 7, 8, 10) math and language arts content. All children’s skill acquisition increased.

2. Caio, F. N., Heejean, G. Y., Finn, H. E., & Ahearn, W. H. (2009). Establishing derived textual control in activity schedules with children with autism. *Journal of Applied Behavior Analysis*, 42(3), 703-709.

This study evaluated whether one-on-one discrete training would serve to transfer the control from activity-schedule pictures to printed words (i.e., derived textual control). Two preschoolers with autism were taught to select pictures and printed words given their dictated names. Following training, participants could respond to printed words by completing the task, match printed words to pictures, and read printed words without explicit training.

3. Eldevik, S., Hastings, R. P., Hughes, J. C., Jahr, E., Eikeseth, S., & Cross, S. (2009). Meta-analysis of early intensive behavioral interventions for children with autism. *Journal of Clinical Child and Adolescent Psychology*, 38(3), 439-450.

This review identified 34 studies using Lovaas' early intensive behavioral intervention model that is often called Lovaas' ABA or DTT. The meta-analysis on young children's IQ and adaptive behavior revealed large and moderate results, respectively. Nine of the 34 studies were controlled designs having either a comparison or control group. The total number of children in the nine studies was 294, 153 were in the early intensive behavioral intervention group. The children's ages ranged from 30.9 to 66.3 months.

4. Schillingsburg, M. A., Kelley, M. E., Roane, H. S., Kisamore, A., & Brown, M. R. (2009). Evaluation and training of yes-no responding across verbal operants. *Journal of Applied Behavior Analysis*, 42(2), 209-223.

Topographically similar verbal responses may be functionally independent forms of operant behavior. For example, saying yes or no may have different functions based on the environmental conditions in effect. The present study sought to teach two children with Asperger Syndrome (ages 3 and 7) yes and no responses. The participants acquired the yes and no responses but did not generalize them.

5. Jennett, H. K., Harris, S. L., & Dehnolino, L. (2008). Discrete trial instruction vs. mand training for teaching children with autism to make requests. *Analysis of Verbal Behavior*, 24, 69-85.

The effectiveness of DTT and mand training were compared in increasing the independent requests of six children with autism. Five of the six participants made more independent requests and required them more quickly using mand training.

6. Dib, N., & Sturmey, P. (2007). Reducing student stereotypy by improving teachers' implementation of discrete-trial teaching. *Journal of Applied Behavior Analysis*, 40, 339-343.

The study evaluated the indirect effects of discrete-trial teaching on three students' stereotypy. Instructions, feedback, modeling, and rehearsal were used to improve implementation of discrete-trial teaching in a private school for children with autism. Findings showed that improvements in accurate teaching were accompanied by systematic decreases in students' levels of stereotypy.

7. Downs, A., Downs, R. C., Johansen, M., & Fossum, M. (2007). Using discrete trial teaching within a public preschool program to facilitate skill development in students with developmental disabilities. *Education & Treatment of Children, 30*(3), 1-27.
The study evaluated the practicality and effectiveness of providing discrete trial training instruction to children with a wide range of developmental disabilities within a public preschool program. Twelve participants were randomly assigned to receive discrete trial training or individual attention in a control condition. The project evaluated the effects of providing discrete trial training on the participants' cognitive, language, behavioral, and social-emotional functioning. Results generally indicated positive changes in adaptive behavior development and social-emotional functioning for students who received discrete trial training.
8. Hilton, J. C., & Seal, B. C. (2007). Brief report: Comparative ABA and DIR trials in twin brothers with autism. *Journal of Autism and Developmental Disorders, 37*, 1197-1201.
Trial interventions in Floor TimeTM and discrete trial training with twin brothers with autism were offered to help the parents choose one of the programs for their sons. Pre- and posttest scores on the Communication and Symbolic Behavior Scales (CSBS) revealed a slight gain in the composite score of the child who received discrete trial training and a slight decrease in the score of the child who received Floor TimeTM.
9. Jones, E. A., Feeley, K. M., & Takacs, J. (2007). Teaching spontaneous responses to young children with autism. *Journal of Applied Behavior Analysis, 40*, 565-570.
This study evaluated the effectiveness of intensive intervention in establishing spontaneous verbal responses of two young children with autism. The intervention involved discrete trial instruction, specific prompts, and error corrections. Data showed positive outcomes of the intervention.
10. Reed, P., Osborne, L. A., & Corness, M. (2007). Brief report: Relative effectiveness of different home-based behavioral approaches to early teaching intervention. *Journal of Autism and Developmental Disorders, 37*, 1815-1821.
The effectiveness of home-based early behavioral interventions for children with autism was studied. A total of 27 participants were sampled. Results indicated that (a) high-intensity behavioral approaches produced greater gains than low-intensity programs; (b) Lovaas and complete application of behavior analysis to approach-type interventions produced the largest gains; and (c) within the high-intensity groups, increased temporal input on the program was not associated with increased gains in the children. Furthermore, the results from clinic-based discrete trial training trials were partially replicated on a home-based sample using children with greater autistic and intellectual impairments.

11. Kroeger, K. A., & Nelson, W. M. (2006). A language programme to increase the verbal production of a child dually diagnosed with Down Syndrome and autism. *Journal of Intellectual Disability Research*, 50, 101-108.
This single-subject experiment examined a program designed to increase the language production and verbal behavior of a 9-year-old boy who had been receiving a 15-hour per week home-based discrete trial training program. Results indicated that language production noticeably increased for each target area after the introduction of the language program and was maintained at a nine-month follow-up session. Thus, a combined treatment approach incorporating direct instruction, natural environment teaching, and incidental teaching was effective in increasing and maintaining responsive and spontaneous speech in a child with Down Syndrome diagnosed with autism.
12. Sigafoos, J., O'Reilly, M., Ma, C. H., Edrisinha, C., Cannella, H., & Lancioni, G. E. (2006). Effects of embedded instruction versus discrete-trial training on self-injury, correct responding, and mood in a child with autism. *Journal of Intellectual & Developmental Disability*, 31, 196-203.
This study compared embedded instruction with discrete-trial training for a 12-year-old boy with autism. Instructional sessions to teach adaptive behaviors were conducted under two conditions: (a) embedded instruction, learning trials were inserted into ongoing activities at a rate of approximately 1.5 per minute; and (b) discrete-trial training, instructional opportunities were incorporated into structured sessions at a rate of four per minute. In both conditions, the system of least prompts was used to teach relevant target responses. Results indicated higher rates of self-injury and fewer correct responses during discrete-trial training. Mood ratings were also lower during discrete-trial training. These findings suggested that although discrete-trial training can be highly effective, it may be preferable to start with embedded instruction when the child presents with self-injurious escape behaviors.
13. Farrell, P., Trigonaki, N., & Webster, D. (2005). An exploratory evaluation of two early intervention programmes for young children with autism. *Educational and Child Psychology*, 22(4), 29-40.
This article discussed the impact of two contrasting early intervention programs, DTT/Lovaas and the Lancashire Under Fives Autism Project (LUFAP) for three young children with autism. The findings suggest that all parents, teachers, and therapists were positive about the impact of both programs. The parents felt supported and were pleased with their children's progress. Teachers, especially those linked to the LUFAP program, were positive about having a child with autism in their school. Data indicated that all the children made progress as measured by the Vineland and the Bayley Scales. However, the rate of progress made by those on the LUFAP program was more encouraging than that of children on the DTT/Lovaas program although this may be attributable to initial group differences. The study included 15 children who were between 3 and 5 years old at the beginning of the study.

14. Grindle, C. F., & Remington, B. (2005). Teaching children with autism when reward is delayed: The effects of two kinds of marking stimuli. *Journal of Autism and Developmental Disorders*, 35, 839-850.

Three children with autism were taught to identify pictures of emotions in response to their spoken names. In the marked-before condition, an instruction encouraged the children to visually orient to the cards before they made their choice response; in the marked-after condition, an attention-eliciting verbal cue (e.g., "Look!") was delivered after both correct and incorrect responses; finally, in the delay condition, the marking cues were omitted. Results showed that performance in the no-cue control was inferior to both the marked-before and marked-after conditions, but the difference between the latter two conditions was not significant.

15. Devlin, S. D., & Harber, M. M. (2004). Collaboration among parents and professionals with discrete trial training in the treatment for autism. *Education and Training in Developmental Disabilities*, 39, 291-300.

This study evaluated the impact of collaborative efforts of parents and school professionals in teaching a 5-year-old boy with autism discrete trial training across settings. Findings showed that young children with AU can obtain significant gains in fine-motor, person/social, language, and perceptual cognitive functioning through the use of DTT occurring 20-24 hours per week across settings.

16. Dillenburger, K., Keenan, M., Gallagher, S., & McElhinney, M. (2004). Parent education and home-based behavior analytic intervention: An examination of parents' perceptions of outcome. *Journal of Intellectual & Developmental Disability*, 29, 119-130.

This study examined parents' perceptions of the outcome of discrete trial training programs. Twenty-two questionnaires were completed by two groups of parents. The first group had just completed an introductory course in applied behavior analysis and was in the early stages of implementing discrete trial training programs with their children. The second group had been involved in education for more than two years. Overall, both groups of parents reported a positive impact. The long-term group reported that they had achieved complex goals with their children, whereas the short-term group reported an immediate positive impact on child and family functioning and parental self-esteem.

17. Grindle, C. F., & Remington, B. (2002). Discrete-trial training for autistic children when reward is delayed: A comparison of conditioned cue value and response marking. *Journal of Applied Behavior Analysis*, 35, 187-190.

Three children with autism (aged 4-8) were taught to identify pictures of objects, and their speed of acquisition of receptive speech skills was compared across two conditions. In the cue-value condition, a compound audiovisual stimulus was presented after correct responses and again when a primary reinforcer was delivered after a 5-second delay. In the response-marking condition, a second stimulus was presented after both correct and incorrect responses, but not prior to the primary reinforcer. In both conditions, primary reinforcement was delayed for five seconds. Although the children learned receptive speech skills in both conditions, acquisition was faster in the cue-value condition.

18. McElwee, J., & Munson, S. (2002). Attaining fluency by a youngster with autism for a beginning listener skill. *Journal of Precision Teaching & Celebration*, 18, 30-32.
The study reported a case of a 7-year-old boy with autism who received a program that used primarily discrete trial training for almost four years. The child was assessed on attaining fluency for the basic listener skill. The study revealed positive outcomes of the intervention. In addition, all retention checks were positive.
19. Charlop-Christy, M. H., & Carpenter, M. H. (2000). Modified incidental teaching sessions: A procedure for parents to increase spontaneous speech in their children with autism. *Journal of Positive Behavior Interventions*, 2, 98-112.
The study compared the efficacy of modified incidental teaching sessions with traditional discrete trial training. Parents of three children with autism were trained to deliver modified incidental teaching, traditional discrete trial, and incidental teaching in their home. The acquisition and generalization of the behavior were measured. Results indicated that modified incidental teaching sessions led to acquisition for all children. By comparison, only one child acquired the behavior with traditional incidental teaching, and two children acquired the behavior with discrete trial. Generalization was documented in the modified incidental teaching sessions, whereas no generalization was found in the incidental teaching and discrete trial conditions.
20. Cummings, A. R., & Williams, W. L. (2000). Visual identity matching and vocal imitation training with children with autism: A surprising finding. *Journal on Developmental Disabilities, Special Issue: Recent Research on the Assessment of Basic Learning Abilities Test*, 7, 109-122.
Five boys with pervasive developmental disorders or autism participated in this study as part of their ongoing weekly in-home, one-to-one, discrete-trial training therapy. For each child, trained therapists conducted daily sessions in a varying order of (a) three-choice identity matching-to-sample using three-dimensional objects, then matching of objects to pictures of objects, then matching of pictures to pictures; (b) Picture Exchange Communication System (PECS); and (c) vocal imitation. Results indicated that all participants learned to match objects to objects, then objects to pictures, and finally pictures to pictures. Shortly after meeting criterion on matching objects to pictures or pictures to pictures, four children learned to respond correctly on PECS. Furthermore, those four children learned to imitate simple sounds within a few sessions after demonstrating mastery of the PECS.
21. Harris, S. L., & Handleman, J. S. (2000). Age and IQ at intake as predictors of placement for young children with autism: A four- to six-year follow-up. *Journal of Autism and Developmental Disorders*, 30, 137-142.
This study examined the predictive power of age and IQ at time of admission to an intensive treatment program using discrete trial training in a 4- to 6-year follow-up of educational placement. Twenty-seven children with autism who were between the ages of 31 and 65 months and had IQs on the Stanford Binet between 35 and 109 at time of admission to a developmental disabilities center were followed up four to six years after they left the

preschool. The results showed that having a higher IQ at intake and being younger were both predictive of being in a general education class after discharge, whereas having a lower IQ and being older at intake were closely related to placement in a special education classroom. In addition, the older children and those with lower IQs showed measurable gains in IQ from treatment.

22. Graff, R. B., Green, G., & Libby, M. E. (1998). Effects of two levels of treatment intensity on a young child with severe disabilities. *Behavioral Interventions*, 13, 21-41.

The study evaluated the effects of two levels of center-based behavioral intervention for a young boy with diagnoses of autism, severe ADHD, bipolar disorder, and severe developmental delay. The child entered school and a residential program that used discrete trial training at age 4. For the first phase of the study, the teacher-to-student ratio was 1:1 and lasted 12 months. Next, the teacher-to-student ratio became 1:2 (the second phase), which continued for nine months. The child's out-of-seat behavior, aberrant behavior, motor imitation, stereotypic responses, matching to sample, and appropriate communication were measured. By the end of the first phase, substantial improvements were documented and medication was discontinued.

23. McEachin, J. J., Smith, T., & Lovaas, O. I. (1993). Long-term outcome for children with autism who received early intensive behavioral treatment. *American Journal of Mental Retardation*, 97, 359-372.

A follow-up of Lovaas' (1987) study was conducted to verify whether behavior treatment may produce long-lasting gains for children with autism. Thirty-eight children with autism participated. Results indicated that (a) children in the experimental group maintained their level of intellectual functioning between the previous assessment at age 7 and the evaluation at a mean age of 13; (b) children in the experimental group displayed significantly higher levels of functioning than did control subjects; and (c) nine children who had been classified as "best-outcome" in the previous study exhibited average intelligence and average levels of adaptive functioning.

24. Miranda-Linné, F., & Melin, L. (1992). Acquisition, generalization, and spontaneous use of color adjectives: A comparison of incidental teaching and traditional discrete-trial procedures for children with autism. *Research in Developmental Disabilities*, 13, 191-210.

This study used incidental teaching and traditional discrete-trial procedures to teach two boys with autism the expressive use of two color adjectives to describe preferred toys and food items. Results showed that traditional discrete-trial teaching was more efficient and produced faster acquisition and, initially, greater generalization. However, by follow-up, the incidental teaching methods resulted in equal retention, greater generalization, and equal or greater spontaneous usage. Together, findings indicate that although it takes longer for children with autism to learn using incidental teaching procedures, once the skills have been acquired, they may be more permanent.

25. Woods, T. S. (1987). Programming common antecedents: A practical strategy for enhancing the generality of learning. *Behavioral Psychotherapy, 15*, 158-180.
Four boys with autism were taught to perform coloring and block-assembly play responses within a discrete-trial training paradigm. Two training strategies that differed with respect to antecedent cueing methods were compared: a naturalistic cueing procedure and a more conventional approach of giving verbal instructions to instigate play responses. Generalization across comparable play “tasks” with only naturalistic cueing was probed. Findings indicated substantially more carryover for all four boys when the antecedent cues for training matched those in the generalization condition.
26. Kayser, J. E., Billingsley, F. F., & Neel, R. S. (1986). A comparison of in-context and traditional instructional approaches: Total task, single trial versus backward chaining, multiple trials. *Journal of the Association for Persons with Severe Handicaps, 11*, 28-38.
The study compared the effects of total-task/single-trial and backward-chaining/multiple-trials instruction in a program teaching eight children with severe disabilities how to make a snack. Findings suggested that total-task/single-trial instruction resulted in superior acquisition of independent steps in the training setting for three of the eight participants and less substantial effects for two participants. For three participants, differences between the two methods were negligible or nonexistent in terms of independent steps. In all cases, instructional time was substantially less for total-task/single-trial instruction than for backward-chaining/multiple-trials instruction. Findings related to generalization were mixed.
27. Cohen, M. (1981). Development of language behavior in an autistic child using total communication. *Exceptional Children, 47*, 379-381.
The study demonstrated the effectiveness of a procedure employing behavior modification techniques (e.g., modeling, stimulus control) in conjunction with a total communication approach in the training of verbal labeling behavior and rudimentary telegraphic speech. It simultaneously decreased echolalia in a 4-year-old girl with autism.
28. Lovaas, O. I., Koegel, R., Simmons, J. Q., & Long, J. S. (1973). Some generalization and follow-up measure on autistic children in behavior therapy. *Journal of Applied Behavior Analysis, 6*, 131-166.
Twenty individuals with autism were treated with behavior therapy. Findings suggested that (a) inappropriate behaviors decreased and appropriate behavior increased, (b) spontaneous social interaction and spontaneous use of language occurred for some of the participants, (c) IQs and social quotients reflected improvement overall – every participant made improvement and skills were maintained during the follow-up.

REFERENCES

- Adcock, A., & Cuvo, A. J. (2009). Enhancing learning for children with autism spectrum disorders in regular education by instructional modifications. *Research in Autism Spectrum Disorders, 3*, 319-328.
- Caio, F. N., Heejean, G. Y., Finn, H. E., & Ahearn, W. H. (2009). Establishing derived textual control in activity schedules with children with autism. *Journal of Applied Behavior Analysis, 42*(3), 703-709.
- Charlop-Christy, M. H., & Carpenter, M. H. (2000). Modified incidental teaching sessions: A procedure for parents to increase spontaneous speech in their children with autism. *Journal of Positive Behavior Interventions, 2*, 98-112.
- Cohen, M. (1981). Development of language behavior in an autistic child using total communication. *Exceptional Children, 47*, 379-381.
- Cummings, A. R., & Williams, W. L. (2000). Visual identity matching and vocal imitation training with children with autism: A surprising finding. *Journal on Developmental Disabilities. Special Issue: Recent Research on the Assessment of Basic Learning Abilities Test, 7*, 109-122.
- Devlin, S. D., & Harber, M. M. (2004). Collaboration among parents and professionals with discrete trial training in the treatment for autism. *Education and Training in Developmental Disabilities, 39*, 291-300.
- Dib, N., & Sturmey, P. (2007). Reducing student stereotypy by improving teachers' implementation of discrete-trial teaching. *Journal of Applied Behavior Analysis, 40*, 339-343.
- Dillenburger, K., Keenan, M., Gallagher, S., & McElhinney, M. (2004). Parent education and home-based behavior analytic intervention: An examination of parents' perceptions of outcome. *Journal of Intellectual & Developmental Disability, 29*, 119-130.
- Downs, A., Downs, R. C., Johansen, M., & Fossum, M. (2007). Using discrete trial teaching within a public preschool program to facilitate skill development in students with developmental disabilities. *Education & Treatment of Children, 30*(3), 1-27.
- Eldevik, S., Hastings, R. P., Hughes, J. C., Jahr, E., Eikeseth, S., & Cross, S. (2009). Meta-analysis of early intensive behavioral interventions for children with autism. *Journal of Clinical Child and Adolescent Psychology, 38*(3), 439-450.
- Farrell, P., Trigonaki, N., & Webster, D. (2005). An exploratory evaluation of two early intervention programmes for young children with autism. *Educational and Child Psychology, 22*, 29-40.

- Graff, R. B., Green, G., & Libby, M. E. (1998). Effects of two levels of treatment intensity on a young child with severe disabilities. *Behavioral Interventions*, 13, 21-41.
- Grindle, C. F., & Remington, B. (2002). Discrete-trial training for autistic children when reward is delayed: A comparison of conditioned cue value and response marking. *Journal of Applied Behavior Analysis*, 35, 187-190.
- Grindle, C. F., & Remington, B. (2005). Teaching children with autism when reward is delayed: The effects of two kinds of marking stimuli. *Journal of Autism and Developmental Disorders*, 35, 839-850.
- Harris, S. L., & Handleman, J. S. (2000). Age and IQ at intake as predictors of placement for young children with autism: A four- to six-year follow-up. *Journal of Autism and Developmental Disorders*, 30, 137-142.
- Hilton, J. C., & Seal, B. C. (2007). Brief report: Comparative ABA and DIR trials in twin brothers with autism. *Journal of Autism and Developmental Disorders*, 37, 1197-1201.
- Jennett, H. K., Harris, S. L., & Dehnolino, L. (2008). Discrete trial instruction vs. mand training for teaching children with autism to make requests. *Analysis of Verbal Behavior*, 24, 69-85.
- Jones, E. A., Feeley, K. M., & Takacs, J. (2007). Teaching spontaneous responses to young children with autism. *Journal of Applied Behavior Analysis*, 40, 565-570.
- Kayser, J. E., Billingsley, F. F., & Neel, R. S. (1986). A comparison of in-context and traditional instructional approaches: Total task, single trial versus backward chaining, multiple trials. *Journal of the Association for Persons with Severe Handicaps*, 11, 28-38.
- Kroeger, K. A., & Nelson, W. M. (2006). A language programme to increase the verbal production of a child dually diagnosed with Down Syndrome and autism. *Journal of Intellectual Disability Research*, 50, 101-108.
- Lovaas, O. I. (1981). *Teaching developmentally disabled children: The me book*. Baltimore: University Park Press.
- Lovaas, O. I. (1987). Behavioral treatment and normal educational and intellectual functioning in young autistic children. *Journal of Consulting and Clinical Psychology*, 55, 3-9.
- Lovaas, O. I., Koegel, R., Simmons, J. Q., & Long, J. S. (1973). Some generalization and follow-up measure on autistic children in behavior therapy. *Journal of Applied Behavior Analysis*, 6, 131-166.
- McEachin, J. J., Smith, T., & Lovaas, O. I. (1993). Long-term outcome for children with autism who received early intensive behavioral treatment. *American Journal of Mental Retardation*, 97, 359-372.

- McElwee, J., & Munson, S. (2002). Attaining fluency by a youngster with autism for a beginning listener skill. *Journal of Precision Teaching & Celebration*, 18, 30-32.
- Miranda-Linné, F., & Melin, L. (1992). Acquisition, generalization, and spontaneous use of color adjectives: A comparison of incidental teaching and traditional discrete-trial procedures for children with autism. *Research in Developmental Disabilities*, 13, 191-210.
- Reed, P., Osborne, L. A., & Corness, M. (2007). Brief report: Relative effectiveness of different home-based behavioral approaches to early teaching intervention. *Journal of Autism and Developmental Disorders*, 37, 1815-1821.
- Schillingsburg, M. A., Kelley, M. E., Roane, H. S., Kisamore, A., & Brown, M. R. (2009). Evaluation and training of yes-no responding across verbal operants. *Journal of Applied Behavior Analysis*, 42(2), 209-223.
- Sigafoos, J., O'Reilly, M., Ma, C. H., Edrisinha, C., Cannella, H., & Lancioni, G. E. (2006). Effects of embedded instruction versus discrete-trial training on self-injury, correct responding, and mood in a child with autism. *Journal of Intellectual & Developmental Disability*, 31, 196-203.
- Smith, T. (2001). Discrete trial training in the treatment of autism. *Focus on Autism and Other Developmental Disabilities*, 16, 86-92.
- Woods, T. S. (1987). Programming common antecedents: A practical strategy for enhancing the generality of learning. *Behavioral Psychotherapy*, 15, 158-180.

RESOURCES AND MATERIALS

- Applied Behavioral Analysis and Discrete Trial Training: <http://www.autismconnectmd.org/education/methods/aba.html>
This links to a brief description of methods and strategies for ABA and DTT.
- A Brief Explanation of Discrete Trial Training: <http://www.iidc.indiana.edu/index.php?pageId=451>
This page is a brief explanation of DTT, including a comparison of DTT and the Lovaas approach.
- Lovaas Institute for Early Intervention: <http://www.lovaas.com/about.php>
This website provides introductory information about the Lovaas approach, which uses DTT.
- Teaching Children with Autism: The Discrete Trial: <http://www.polyxo.com/discretetrial>
This resource presents the scope of DTT and provides a basic example.

GENERAL RESOURCES

- Autism Internet Modules (AIM) www.autisminternetmodules.org
The Autism Internet Modules were developed with one aim in mind: to make comprehensive, up-to-date, and usable information on autism accessible and applicable to educators, other professionals, and families who support individuals with autism spectrum disorders (ASD). Written by experts from across the U.S., all online modules are free, and are designed to promote understanding of, respect for, and equality of persons with ASD.
Current modules are:
 - Assessment for Identification
 - Home Base
 - Peer-Mediated Instruction and Intervention (PMII)
 - Picture Exchange Communication System (PECS)
 - Pivotal Response Training (PRT)
 - Preparing Individuals for Employment
 - Reinforcement
 - Restricted Patterns of Behavior, Interests, and Activities
 - Self-Management
 - Social Supports for Transition-Aged Individuals
 - Structured Teaching
 - Structured Work Systems and Activity Organization
 - Supporting Successful Completion of Homework
 - The Incredible 5-Point Scale
 - Time Delay
 - Transitioning Between Activities
 - Visual Supports
- Interactive Collaborative Autism Network (iCAN) <http://www.autismnetwork.org>
iCAN offers free online instructional modules on autism spectrum disorder (ASD). Modules have been developed in these areas:
 - Characteristics
 - Assessment
 - Academic Interventions
 - Behavioral Interventions
 - Communication Interventions
 - Environmental Interventions
 - Social Interventions
- Indiana Resource Center for Autism (IRCA) <http://www.iidc.indiana.edu/irca/fmain1.html>
The Indiana Resource Center for Autism staff's efforts are focused on providing communities, organizations, agencies, and families with the knowledge and skills to support children and adults in typical early intervention, school, community, work, and home settings.
 - IRCA Articles <http://www.iidc.indiana.edu/irca/ftrainpapers.html>
 - IRCA Modules <http://www.iidc.indiana.edu/irca/fmodules.html>

- Texas Statewide Leadership for Autism www.txautism.net
The Texas Statewide Leadership for Autism in conjunction with the network of Texas Education Service center with a grant from the Texas Education Agency has developed a series of free online courses in autism. Please check the training page, www.txautism.net/training.html, for update lists of courses, course numbers and registration information. Current courses include the following:
 - Autism 101: Top Ten Pieces to the Puzzle
 - Autismo 101: Las 10 piezas principales del rompecabezas
 - Asperger Syndrome 101 Online
 - Asperger Syndrome 101 Online
 - Navigating the Social Maze: Supports & Interventions for Individuals with Autism Spectrum Disorders
 - Communication: The Power of Communication for Individuals with Autism Spectrum Disorders
 - Communication: The Power of Communication for Individuals with Autism Spectrum Disorders

ERRORLESS LEARNING

CHARACTERISTICS OVERVIEW CHART

Verbal Skills	Grade Levels	Cognitive Level	Areas Addressed
☐ Nonverbal	☒ PK	☒ Classic	☒ (Pre)Academic/Cognitive/Academic
☒ Mixed	☒ Elementary	☐ High Functioning	☐ Adaptive Behavior/ Daily Living
☒ Verbal	☐ Middle/High		☐ Behavior ☒ Communication/Speech ☐ Social/Emotional

BRIEF INTRODUCTION

Children with autism (AU) share common characteristics in learning. For example, they adhere rigidly to routines and tend to over-select and over-generalize responses to failure or novel tasks. Therefore, errorless learning, which limits an incorrect response in a learning situation, is ideal for this group of students.

DESCRIPTION

Errorless learning, a procedure introduced by Terrace (1963), is a type of discrimination learning that decreases or eliminates the opportunity for incorrect choice selection, therefore maximizing the possibility of a correct response. Simply put, errorless learning allows learning to occur with few or no negative stimuli. The theory behind errorless learning is that error responses have negative effects, especially for children with autism, given their rigid adherence to rules (Green, 1996; Smith, 2001; Smith, Iwata, Goh, & Shore, 1995).

Errorless learning offers the following benefits:

- Minimizes the number of errors
- Increases overall time available for instruction
- Reduces the likelihood that errors will be repeated in future trials
- Reduces frustration and the occurrence of inappropriate emotional behaviors by increasing opportunities for reinforcement

In errorless learning, children only learn the correct skill. That is, the teacher teaches in such a manner that students do not make any mistakes. As a result, they do not learn an incorrect skill that will have to be corrected or re-taught.

STEPS

Guidelines for using errorless learning are as follows:

1. Identify and teach the child the desired behavior.
2. Identify prompts that will ensure success.
3. Have the child begin to perform the response.
4. Provide prompts to make sure the child performs the desired behavior correctly.
5. If behavior/response is incorrect, increase prompt to make the child successful.
6. Repeat the trial several times until the child appears to be able to demonstrate the desired behavior correctly and independently.
7. Following a specified number of non-prompted behavior, conduct a trial to assess the child's correct or incorrect learned behavior.
8. Finish the lesson on a successful trial with appropriate reinforcement.
9. Fade or decrease prompting as soon as indicated by data collection.

BRIEF EXAMPLE

Ms. Cooper utilized errorless learning techniques in teaching John, a 3-year-old boy with autism, to recognize his body parts. She asked John to touch the body part that she named. At first, Ms. Cooper provided a full prompt by taking John's hand and touching the correct body part. She gave John a small cookie as a reinforcer whenever he finished the task.

After three trials, Ms. Cooper faded the prompt by merely lifting John's hand toward the correct body part. When John successfully performed the task, he received a small cookie. When John failed to perform the task, Ms. Cooper prompted him through the task and provided a reinforcer. Gradually, Ms. Cooper faded prompts. After several trials, John could successfully perform the task with no prompts.

SUMMARY

Errorless learning is a set of teaching techniques designed to reduce incorrect responses as the child gains mastery of a novel task. It has been contrasted with trial and error learning in which the child attempts a task and then benefits from feedback. This strategy is an effective way in which to teach a variety of skills to individuals with autism.

RESEARCH TABLE

Number of Studies	Ages (year)	Sample Size	Area(s) Addressed	Outcome
1	4-7	3	Word acquisition	+

STUDIES CITED IN RESEARCH TABLE

- I. Goldsmith, T. R., LeBlanc, L. A., & Sautter, R. A. (2007). Teaching intraverbal behavior to children with autism. *Research in Autism Spectrum Disorders, 1*, 1-13.
In this study, three children with autism were successfully taught to name items associated with preselected categories (e.g., "What are some colors?") with limited generalization to a fourth, non-targeted category. Limited maintenance of skills was found.

REFERENCES

- Goldsmith, T. R., LeBlanc, L. A., & Sautter, R. A. (2007). Teaching intraverbal behavior to children with autism. *Research in Autism Spectrum Disorders, 1*, 1-13.
- Green, G. (1996). Behavioral intervention for autism. In C. Maurice, G. Green, & S. C. Luce (Eds.), *Behavioral interventions for young children with autism* (pp. 29-42). Austin, TX: Pro-Ed.
- Smith, T. (2001). Discrete trial training in the treatment of autism. *Focus on Autism and Other Developmental Disabilities, 16*, 86-92.
- Smith, R. G., Iwata, B. A., Goh, H., & Shore, B. A. (1995). Analysis of establishing operations for self-injury maintained by escape. *Journal of Applied Behavior Analysis, 28*, 515-535.
- Terrace, H. S. (1963). Discrimination learning with and without "error." *Journal of the Experimental Analysis of Behavior 6*, 1-27.

RESOURCES AND MATERIALS

- LEARNet Problem-Solving System and
Resource: http://www.projectlearn.net/tutorials/errorless_learning.html
This URL links the user to the errorless learning tutorial of the Brain Injury Association of New York. It includes a video illustration and several practical application tips.
- Errorless Learning/Teaching. ASAT (Association for Science in Autism Treatment): <http://www.asatonline.org/intervention/procedures/errorless.htm>
This site provides a research summary and recommendation on this intervention.

GENERAL RESOURCES

- Autism Internet Modules (AIM) www.autisminternetmodules.org
The Autism Internet Modules were developed with one aim in mind: to make comprehensive, up-to-date, and usable information on autism accessible and applicable to educators, other professionals, and families who support individuals with autism spectrum disorders (ASD). Written by experts from across the U.S., all online modules are free, and are

designed to promote understanding of, respect for, and equality of persons with ASD.

Current modules are:

- Assessment for Identification
 - Home Base
 - Peer-Mediated Instruction and Intervention (PMII)
 - Picture Exchange Communication System (PECS)
 - Pivotal Response Training (PRT)
 - Preparing Individuals for Employment
 - Reinforcement
 - Restricted Patterns of Behavior, Interests, and Activities
 - Self-Management
 - Social Supports for Transition-Aged Individuals
 - Structured Teaching
 - Structured Work Systems and Activity Organization
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From Resource Room to the Real World

Facilitating Generalization of Intervention Outcomes

Jonathan L. Burt and Todd Whitney

Eddie is a sixth-grade student with a specific learning disability in reading. During his most recent annual individualized education program (IEP) meeting, his general education teacher reported that Eddie was not keeping up with his independent classwork. She stated that in social studies Eddie often will sit quietly at his desk and pretend to complete the assignment by copying irrelevant text from his book. Similarly in math and science, he will write random answers to problems without engaging the content as instructed. Even though Eddie has been taught how to request help by his special education teacher, he was not transferring this skill to his general education classes. When asked why, Eddie stated that he doesn't know when or how to get help from his teachers—they always look so busy. Eddie's special education teacher, Mrs. Claire, wonders why Eddie raises his hand for help in the resource classroom but not in his general education classes. She asked herself, "What can I do to encourage Eddie to transfer this skill to where it matters most?"

Many special education teachers have encountered challenges similar to Mrs. Claire's situation with Eddie. Teaching a skill within the highly controlled conditions of one setting and promoting the transfer of that skill to a less-structured setting may seem like an unattainable goal. Yet this is the challenge presented by many IEP objectives that involve generalization of a skill acquired in a resource room to be applied outside of the resource room. In fact, this is a challenge for any learning task where one environment is selected for initial skill acquisition with the expectation that the behavior occurs elsewhere. The ability to generalize skills across settings is essential at all stages of life for nearly every learned skill.

Although generalization, or generality as it is sometimes called, is one of the most important outcomes of applied behavioral interventions (Cooper, Heron, & Heward, 2007), it frequently fails to occur naturally following initial intervention efforts. In

other words, generalization is not guaranteed. The challenges associated with the lack of generalization were first articulated by Baer, Wolf, and Risley (1968); fortunately, however, progress has been made toward the development of systematic generalization promotion strategies (Osnes & Lieblin, 2003).

The ability to generalize skills across settings is essential at all stages of life for nearly every learned skill.

In 1977, Stokes and Baer published a review of best practices regarding generalization of applied behavioral intervention outcomes. In this seminal review, they defined generalization as the extension of behavior change across "different, nontraining conditions (i.e., across subjects, settings, people, behaviors, and/or time) without the scheduling of the same events in those conditions as had been scheduled in the training conditions" (Stokes & Baer, 1977, p. 350; see Table 1 for definitions and examples of these domains). In other words, generalization is observed when behavior change accomplished under certain conditions, targeting a specific behavior, and with specific students is subsequently observed in different settings (e.g., from a resource room setting to a general education classroom), with novel people (e.g., a student not directly involved in training), with new behaviors (i.e., a behavior not directly targeted in training), and at a different time (i.e., maintenance or follow-up).

Despite a lack of research demonstrating precisely how behaviors generalize from one setting to the next (Osnes & Lieblin, 2013), a recent increase in the publication of studies including generalization measures as dependent variables affirms the importance of this outcome (Swan, Carper, & Kendall, 2015). This research underscores the importance of using specific generalization-promotion strategies as effective practice for both academic and behavioral interventions.

Specifically, strategies have been used to generalize completion of academic tasks (Mesmer, Duhon, & Dodson, 2007) as well as promote early literacy skills (Silber & Martens, 2010) and oral reading fluency (Duhon, House, Poncy, Hastings, & McClurg, 2010). Programming for generalization has also been effective in promoting social

interactions (Deitchman, Reeve, Reeve, & Progar, 2010), task accuracy and independence (Hume, Plavnick, & Odom, 2012), and conversation skills (Spencer & Higbee, 2012). Furthermore, the Council for Exceptional Children and the Collaboration for Effective Educator Development, Accountability, and Reform (CEEDAR) Center published a collection of 22 high-leverage practices (HLPs) that are described as essential to promoting improved outcomes for students with disabilities (McLeskey et al., 2017). One of these practices is to "teach students to maintain and generalize new learning across time and settings" (HLP 21; McLeskey et al., 2017).

Generalization Programming Strategies

Stokes and Osnes (1989) organized the specific strategies first summarized by Stokes and Baer (1977) into three generalization promotion categories: (a) train diversely, (b) exploit functional contingencies, and (c) incorporate functional mediators (Table 2).

Train Diversely

Teaching procedures that call for a narrow range of specific responses are not likely to generalize to new environments. Similarly, an instructional setting, if too repetitive and predictable, may result in overly tight instructional control. In other words, a student is more likely to perform a learned behavior in generalization settings only if that setting closely mirrors the initial

Table 1. The Domains of Generalization

Domain	Definition	Example
Setting or situation	Appropriate responding in new or novel environmental conditions	- Teaching students a strategy to answer mathematics word problems in the resource room, which is then applied during mathematics class - Teaching students to say “hello” to you, and then students starts saying “hello” to other teachers
Response	Variety of behaviors that are altered or changed as a result of direct training of a specific response	- Praising students for writing their names neatly at the top of paper, and the handwriting on the rest of the assignment improves - Reminding students as they enter the bathroom to wash their hands, and in the process, they also throw away their paper towels, which had not been occurring previously
Time	Student maintains correct responding when parts or all of the instructional supports are removed (i.e., maintenance) or following extended periods of time after the conclusion of instruction (i.e., follow-up)	- Students uses problem-solving mnemonic even though the poster with a visual of the mnemonic has been removed - Hand-raising behavior in the general education classroom is continued after verbal reminders have been removed
Participants	Students not directly involved in the training interactions demonstrate acquisition of the skill	- After seeing the poster, other students begin using a mnemonic that was taught to the target student - Other students start raising their hands after hearing the target student being reinforced for hand raising

teaching conditions. For example, Eddie’s initial advocacy training in the resource room consisted of Mrs. Claire presenting independent worksheets, followed by her modeling hand raising, then prompting Eddie to do the same. Each time Eddie raised his hand, Mrs. Claire immediately attended to his question and offered help or feedback. Eddie quickly mastered this skill by correctly responding in five consecutive trials.

Despite a benefit of quick skill acquisition, repetitive instruction in this case likely produced overly restricted behavior patterns. When initially learning the skill, Eddie practiced hand raising only when given a worksheet during independent work. In general education classrooms, however, hand raising is but one of many ways to obtain assistance, and independent worksheets compose a small percentage of the activities with which Eddie will likely need assistance. As discussed in detail in the next section, Eddie’s technique for obtaining assistance may be expanded by varying the type of presentation materials and instructional methods used

and reinforcing a wide range of response forms.

Use a variety of stimulus and response examples. Stimulus examples are the prompts that occur before the behavior and response examples are the various forms of behavior (i.e., what the behavior looks like) targeted in instruction. To maximize the probability of generalization, it is important to incorporate a variety of stimulus and response examples. The method of general case instruction developed by Horner and colleagues in the 1980s helps to formalize this process (Horner & McDonald, 1982; Sprague & Horner, 1984). General case instruction involves a systematic assessment of the various stimulus arrangements (i.e., range of activities, circumstances, responses) likely encountered in a student’s natural environment and presentation of a sample of that range during instruction (see Table 3 for an outline of recommended procedures). It is not practical, or in many cases even possible, to incorporate every stimulus example a student will encounter in

natural settings; therefore, the goal of general case instruction is to efficiently teach a skill by minimizing the number of examples needed while maximizing the probability of correct responding when confronted with new but similar problems. For example, Mrs. Claire trained Eddie to raise his hand for assistance while completing a worksheet during independent work. However, independent work involves a wide range of activities involving a variety of materials (e.g., textbooks, computers, manipulatives). Assessing the general circumstances in which this skill is necessary would help Mrs. Claire determine the full range of scenarios in which Eddie would likely need to raise his hand for assistance.

When implementing general case instruction, it is also important to consider the range of situations in which a new response would not be appropriate (i.e., overgeneralization). For example, it is appropriate for Eddie to raise his hand for assistance during independent work with tasks that are challenging for him; however, raising his hand in the middle of group instruction for unrelated

Table 2. Generalization Program Strategies

Category	Strategies	How to incorporate	Ms. Claire's example
Train diversely	Use sufficient stimulus and response exemplars	Use a variety of antecedent cues and response topographies that sample the range of those employed in natural settings	Mrs. Claire considered all relevant settings and situations in which Eddie would need to request help.
	Make antecedents and consequences less discriminable	Avoid predictable schedules of reinforcement to make it harder for student to determine when consequences for actions will be available	Mrs. Claire taught Eddie how to raise hand, wait to be called on, and try again later if teacher was unavailable.
Exploit functional contingencies	Contact natural contingencies	Give highest priority to behaviors and skills that will likely be reinforced in the general education classroom	Mrs. Claire verified hand raising was the primary means of gaining attention in the general education classroom.
	Reinforce generalization	Use preference assessments, behavior contracts, daily progress reports, or token economies to reinforce target behavior. ^a	Mrs. Claire used a preference assessment to determine Eddie would like extra computer time.
	Modify maladaptive consequences	Work with general education teacher on direct interventions to modify counterproductive consequences in general education classroom	Mrs. Claire learned some students get made fun of for answering too many questions. She asked general education teacher to monitor.
Incorporate functional mediators	Incorporate common environmental stimuli	Have instructional setting be as similar in form and function to the intended generalization setting as possible	Mrs. Claire scheduled time in the library to train on computers and asked Mr. Harris to model traffic light signal system.
	Incorporate self-mediated physical, verbal, and covert stimuli	Use self-management interventions that prompt behavior in a tangible (e.g., checklists or timers) or covert (e.g., mnemonics or catch phrases) manner.	Mrs. Claire taught Eddie to self-monitor and record each attempt to recruit teacher attention.

a. Reinforcement systems should be carefully and systematically faded.

questions (e.g., asking about whether or not he is supposed to go to after-school care) is not appropriate. The possibility of overgeneralization challenges the teacher to consider how far a behavior must generalize before going too far. There is no formula for determining the scope of generalization in every situation, as correct responding is often determined by the individual context of a particular setting. It is difficult, if not impossible, to envision every scenario where a skill is and is not applicable.

However, teachers are encouraged to do the best they can to sample the range of examples and nonexamples in original instruction. If a circumstance arises in which overgeneralization occurs, teachers must simply address it by incorporating the scenario in another round of instruction.

Mrs. Claire resolved to help Eddie learn to recruit teacher feedback and assistance, but she struggled with where to begin. She vaguely remembered

something about transferring behavior change across settings, and when she consulted her classroom management textbook, she rediscovered the chapter on generalization. Bingo! Step 1: Train diversely. To begin, Mrs. Claire considered all the relevant settings and situations in which Eddie would need to request help as well as the various acceptable and effective ways students get their teacher's attention. To do this, she created a form on which Eddie's general education and related arts

Table 3. General Case Instruction Procedures

Step	Eddie's example
1. Define the range of generalization settings	Eddie will use self-advocacy skills (i.e., asking for teacher assistance) in general education classrooms, including special areas, during independent activities
2. List all relevant stimulus activities, setting circumstances, and response examples	<i>Stimulus</i> activities: Independent activities, including worksheets, computer or tablet assignments, teacher-assigned tasks with textbooks, and projects involving tangibles or manipulatives <i>Setting circumstances</i>: Teacher available, teacher busy helping other students, teacher busy with other tasks <i>Response</i>: Hand raising, vocal interjection, stand and wait, cue cards
3. List potential areas for overgeneralization	Hand raising in Mr. Harris's room is forbidden
4. Select and sequence teaching examples	- Hand raising during independent worksheet and teacher is available - Vocal interjection during teacher-assigned task with textbook and teacher is busy with other tasks (e.g., filing papers) - Stand and wait during computer assignment and teacher is busy with other students - Cue card during manipulative assignment and teacher is available
5. Deliver instruction with progress monitoring	- Outline rationale and steps for obtaining assistance in general education classes - Model technique for each stimulus and setting example - Provide opportunities for guided practice - Give positive or corrective feedback based on student performance - Record student performance in table or graph format

teachers could list the independent activities their students might be completing over the next month. Also included on this form was a checklist of various behaviors students typically use to obtain teacher attention (i.e., hand raising, seat signals, shouts, other). Finally, Mrs. Claire included a comment section where teachers could identify their “pet peeves” regarding teacher attention recruitment in order to plan for and hopefully avoid any instances of overgeneralization.

With this feedback, Mrs. Claire learned that Eddie will primarily encounter independent or small-group assignments in his general education classroom settings. However, she noted that Eddie's social studies class will be in the library using laptops for the next 2 weeks. Therefore, Mrs. Claire planned to include the library with computers among her more standard stimulus examples. Mrs. Claire also noticed that hand raising was the preferred method of teacher attention recruitment among Eddie's teachers—all except for Mr. Harris. Mr. Harris discourages students from calling

out his name in class; therefore he has prohibited hand raising during independent work and replaced it with traffic light seat signals (i.e., green card = I'm fine; yellow = I need help or feedback; red = I need a break). Mrs. Claire identified hand raising in Mr. Harris's class as an example of overgeneralization and incorporated this into her lesson plan.

Make antecedents and consequences less discriminable. A second reason teaching procedures may fail to produce generalization is that students often attempt a strategy only if they are aware that reinforcement for that behavior is available and likely to occur. For example, during initial attempts to teach Eddie hand-raising skills in the resource room, he raised his hand for every trial and received immediate feedback each time. Therefore, Eddie knew every time he raised his hand, it would result in teacher assistance. In other words, during initial training, the positive consequence was highly likely and highly discriminable. Eddie's general

education teachers, by contrast, frequently have 30 or more students at a time. It may not be possible for his teachers to attend to each student every time there is a question. Even if Eddie had attempted to raise his hand, there is a good chance a teacher might not have noticed or otherwise could not have immediately attended to him. Therefore, Eddie must be taught to persist under less-than-optimal and uncertain reinforcement conditions. If consequences are unpredictable in the generalization setting, they need to be unpredictable in the training setting to ensure a higher probability of maintained responding.

Next, Mrs. Claire considered how discriminable antecedents and consequences may be interfering with the transfer of hand raising between the resource and general education settings. During training, Eddie successfully obtained teacher attention and assistance each time he raised his hand. Mrs. Claire realized, however, that this may be setting

Eddie up for failure as students in general education classes may have to try several times to obtain teacher assistance. Mrs. Claire decided to teach Eddie how to raise his hand, wait to be called on, and try again later if a teacher was unavailable. To do this, Mrs. Claire would teach Eddie to count slowly to 30 while he raised his hand. If after 30 seconds, the teacher did not notice Eddie's signal, Eddie would lower his hand and try again later.

Exploit Functional Contingencies

A second category of generalization programming strategies involves direct management of the environmental contingencies related to the behavior. A contingency is the when-if-then relation that exists between the behavior and the antecedents (i.e., events that precede a behavior) and consequences (i.e., events that follow a behavior). Many instances of behavior are seen as functionally related to these environmental contexts. Therefore, to maximize the probability that a newly acquired behavior may be demonstrated under different circumstances, teachers should be aware of the circumstances in which the target behavior is expected to occur in the generalization setting. If the generalization setting is not likely to reinforce the behavior, consistently reinforce a competing behavior, or punish a newly acquired behavior, no amount of training is going to transfer the behavior change to the less-than-supportive generalization setting. Three strategies that may help a teacher address these challenges are to (a) train behaviors likely to contact natural contingencies of reinforcement, (b) artificially reinforce occurrences of generalization, and (c) modify maladaptive consequences.

Contact natural consequences.

Promoting generalization begins with the selection of a skill or behavior to be targeted for acquisition. Before any consideration of *how* a skill should be taught, teachers should first consider *what* is being taught and whether the skill meets the needs of the student and will be reinforced naturally. As any

advanced algebra teacher can testify, students have a natural inclination to

Before any consideration of *how* a skill should be taught, teachers should first consider *what* is being taught and whether the skill meets the needs of the student and will be reinforced naturally.

want to know how a skill is going to benefit them in the future. In other words, students seem to be asking, "What's in it for me?" A best practice in promoting generalization includes teaching behaviors only if they are likely to contact reinforcement in natural environments, a principle known as the relevance-of-behavior rule (Ayllon & Azrin, 1968). Therefore, highest priority should be given to behaviors and skills that will likely produce positive results in the generalization setting.

Mrs. Claire considered the relevance-of-behavior rule regarding the target behavior (i.e., hand raising). She asked herself, "Is this required of other students in general education classes? Does hand raising actually result in teacher assistance in Eddie's classes?" Mrs. Claire scheduled several observations in Eddie's general education classrooms to verify the use of hand raising by most students as a primary means of obtaining teacher assistance. This was confirmed by her observations.

Reinforce generalization. In some instances, a student must continue to behave as instructed even if the natural environment does not provide sufficient reinforcement. In such cases, it may be necessary to motivate the student through the use of artificial reinforcement. Preference assessments, behavior contracts, daily progress reports, and token economies are a few of a wide variety of techniques behavior analysts and educators use to encourage application of a skill. These strategies, although effective at increasing motivation, must be used with discretion. Any artificial

reinforcement system should be carefully and systematically faded,

when possible, to avoid dependence on the system at the expense of the natural reinforcement.

Mrs. Claire determined Eddie may benefit from additional incentives beyond the natural consequences of receiving help in class to build momentum with using this skill independently. She administered a preference assessment and determined that Eddie would like additional time on the computer. Mrs. Claire let Eddie know that he could earn a reward of 5 minutes of computer time per question asked in class.

Modify maladaptive consequences.

Finally, teachers should consider behaviors that are already being reinforced in the generalization setting that compete with the behavior targeted in training and consequences that either reduce, eliminate, or punish the target behavior. Many times, direct intervention in generalization settings to modify counterproductive consequences is necessary.

From conversations with Eddie, Mrs. Claire learned that some kids get made fun of for asking too many questions in class. Therefore, Mrs. Claire asked Eddie's teachers to monitor student reactions to Eddie's participation and assist as needed to address this maladaptive consequence.

Incorporate functional mediators

A third and final category of generalization programming involves the use of mediation strategies. Despite strong instructional design and delivery with carefully planned behavior management systems in place, generalized responding may still fail to

occur. Mediation strategies may be needed to bridge the gap between training and generalization settings. Two specific procedures receive special attention among the generalization literature: (a) incorporate common environmental stimuli and (b) incorporate self-mediated physical, verbal, and covert stimuli (Stokes & Osnes, 1989).

Incorporate common environmental stimuli. For academic skills, preprogrammed instructional materials exist to assist teachers with generalization training (e.g., Direct Instruction; Engelmann & Carnine, 1982). However, packaged social-skills programs are less helpful and often fail to produce the intended results due to the complexity and unique circumstances of social skills (Maag, 2005). Instead,

the traffic light signal system. Both techniques, Mrs. Claire hoped, would help to incorporate common stimuli between the training and generalization setting

Incorporate self-mediated physical, verbal, and covert stimuli. Finally, self-management interventions that include components such as self-mediated physical, verbal, or covert stimuli may be needed to serve as remote prompts to remind a student about what to do. When teaching academic objectives, teachers may list the steps required to solve a problem and allow the student to reference these supports when learning to perform a skill. The same procedures are helpful with social behaviors. Something as simple as a reminder card taped to a student's desk in both training and

steps required to obtain assistance from a teacher, with a starred reminder about Mr. Harris' class. In addition, instead of the teacher recording Eddie's performance, Eddie was taught to self-monitor and record each attempt to recruit teacher attention with an open circle. When successful, Eddie is to shade in the circle and present the results to the teacher at the end of class for verification and feedback.

Following her renewed efforts to teach Eddie self-advocacy skills, Mrs. Claire sent him off to his first class: science with Mr. Harris. After 50 long minutes, Eddie returned to Mrs. Claire's class to check in. With his head down and self-monitoring form squashed in his hand, Eddie approached Mrs. Claire. "How'd it go?" she asked nervously. Eddie smiled and handed her the form. "Can I use the swivel chair for my computer time?" he asked. Mrs. Claire saw three open circles and one closed circle next to Mr. Harris's initials followed by about 30 exclamation points. It was only then Mrs. Claire realized the reward of 5 min of computer time for every hand raise was way too much time!

The ability of teachers' efforts to produce generalization outcomes is crucial to the development of independent learners capable of succeeding in the real world.

teachers must design lesson plans and obtain materials independently. To guide efforts toward this end, it is recommended that the training setting be as similar in form and function to the intended generalization setting as possible. This may require altering the physical arrangement of the training setting to mirror that of the generalization setting. Incorporating common personnel among the settings may help, as well. If a general education teacher or students from the general education class are able to participate in training, all the better. The more features the training setting and generalization setting have in common, the greater the probability that generalization will occur.

Before Mrs. Claire introduced Eddie to the new training protocol, she scheduled time in the library to conduct a training session with the laptops they will be using in social studies. Additionally, she recruited Mr. Harris to volunteer 10 minutes of his planning period to model

generalization settings may be all that is necessary to prompt a student to engage in a behavior across both settings. More elaborate systems may involve the use of electronic timers, such as a MotivAider or a smartphone application, to cue the student to implement the behavior as trained. Self-management checklists and mnemonic memory aids are additional interventions designed to facilitate independent responding apart from initial training settings. The common theme among these mediation strategies is the transfer of the role of the teacher to the student. Whatever strategies are used in the training setting to prompt and reinforce a student to complete a task should be outsourced in a tangible (e.g., checklists or timers) or covert (e.g., mnemonics or catch phrases) manner to increase the probability of generalization.

Mrs. Claire modified Eddie's daily progress report to include a list of the

Conclusion

Programming for generalization is perhaps one of the most challenging tasks for behavior interventionists (Hayes, Rincover, & Solnick, 1980) because much of the behavioral process involved with generalization is unknown (Osnes & Lieblein, 2003). Although answers to the question "How does generalization happen?" remain elusive among the research community, increasingly, researchers have demonstrated positive generalization outcomes using strategies similar to those outlined in this article (Swan et al., 2015). In addition to this basic menu of strategies, the following recommendations serve as a summary to guide practitioners' efforts toward generalization promotion (Cooper et al., 2007, pp. 650–651):

1. *Minimize the need for generalization as much as possible.* Access to a

student's least restrictive environment is not just a federal requirement; it is best practice for generalization promotion.

Conducting instruction in the environments within which a student is expected to perform the skill eliminates the need for generalization promotion strategies in the first place.

2. *Conduct generalization probes before, during, and after instruction.* Close, continual contact with relevant outcome data (Bushell & Baer, 1994) is a hallmark of applied behavior analysis. Generalization, as with any academic or social skill objective, requires consistent formative data collection to guide ongoing efforts and ultimately evaluate the status of the objective.
3. *Involve significant others whenever possible.* Incorporating parents, teachers, staff, or other students. From the generalization settings in initial instruction serves much the same purpose as programming common stimuli. The most relevant controlling stimuli from a generalization setting are often the people with whom our students interact. Bringing those individuals to the instructional setting and incorporating them into training not only serves as common stimuli for the student but also teaches others to be more receptive to the target behavior.
4. *Promote generalization with the least intrusive and least costly methods possible.* Avoid unnecessary use of artificial reinforcement. If a behavior is to be functional in natural settings, it must be brought under the control of natural schedules of reinforcement. Whatever additional reinforcers are used to initially teach a skill must eventually be faded.
5. *Contrive intervention approaches as needed to achieve important generalized outcomes.* With the previous points in mind, if generalization promotion is the ultimate goal, then teachers should

employ whatever strategies are necessary within the categories of *train diversely*, *exploit functional contingencies*, and *mediate generalization*.

Although generalization promotion is by no means easy, it is well worth the effort. The ability of teachers' efforts to produce generalization outcomes is crucial to the development of independent learners capable of succeeding in the real world.

References

- Ayllon, T., & Azrin, N. (1968). *The token economy: A motivational system for therapy and rehabilitation*. New York, NY: Appleton-Century-Crofts.
- Baer, D. M., Wolf, M. M., & Risley, T. R. (1968). Some current dimensions of applied behavior analysis. *Journal of Applied Behavior Analysis*, 1, 91–97. doi:10.1901/jaba.1968.1-91
- Bushell, D., & Baer, D. (1994). Measurably superior instruction means close, continual contact with the relevant outcome data: Revolutionary. In R. Gardner, D. M. Sainato, J. O. Cooper, T. E. Heron, W. L. Heward, J. Eshleman, & T. A. Grossi (Eds.), *Behavior analysis in education: Focus on measurably superior instruction* (pp. 3–10). Monterey, CA: Brooks/Cole.
- Cooper, J. O., Heron, T. E., & Heward, W. L. (2007). *Applied behavior analysis* (2nd ed.). Upper Saddle River, NJ: Pearson.
- Deitchman, C., Reeve, S. A., Reeve, K. F., & Progar, P. R. (2010). Incorporating video feedback into self-management training to promote generalization of social initiations by children with autism. *Education and Treatment of Children*, 33, 475–488. doi:10.1353/etc.0.0102
- Duhon, G. J., House, S. E., Poncy, B. C., Hastings, K. W., & McClurg, S. C. (2010). An examination of two techniques for promoting response generalization of early literacy skills. *Journal of Behavioral Education*, 19, 62–75. doi:10.1007/s10864-010-9097-2
- Engelmann, S., & Carnine, D. (1982). *Theory of instruction: Principles and applications*. New York, NY: Irvington.
- Hayes, S. C., Rincover, A., & Solnick, J. V. (1980). The technical drift of applied behavior analysis. *Journal of Applied Behavior Analysis*, 13, 275–285. doi:10.1901/jaba.1980.13-275
- Horner, R. H., & McDonald, R. S. (1982). Comparison of single instance and general case instruction in teaching a generalized vocational skill. *Research and Practice for Persons With Severe Disabilities*, 7, 7–20. doi:10.1177/154079698200700302
- Hume, K., Plavnick, J., & Odom, S. L. (2012). Promoting task accuracy and independence in students with autism across educational setting through the use of individual work systems. *Journal of Autism and Developmental Disorders*, 42, 2084–2099. doi:10.1007/s10803-012-1457-4
- Maag, J. W. (2005). Social skills training for youth with emotional and behavioral disorders and learning disabilities: Problems, conclusions, and suggestions. *Exceptionality*, 13, 155–172. doi:10.1207/s15327035ex1303_2
- McLeskey, J., Barringer, M.-D., Billingsley, B., Brownell, M. Jackson, D., Kennedy, M., . . . Ziegler, D. (2017). *High-leverage practices in special education*. Arlington, VA: Council for Exceptional Children and CEEDAR Center.
- Mesmer, E. M., Duhon, G. J., & Dodson, K. G. (2007). The effects of programming common stimuli for enhancing stimulus generalization of academic behavior. *Journal of Applied Behavior Analysis*, 40, 553–557. doi:10.1901/jaba.2007.40-553
- Osnes, P. G., & Lieblein, T. (2003). An explicit technology of generalization. *Behavior Analyst Today*, 3, 364–374. doi:10.1037/h0099994
- Silber, J. M., & Martens, B. K. (2010). Programming for the generalization of oral reading fluency: Repeated readings of entire text versus multiple exemplars. *Journal of Behavioral Education*, 19, 30–46. doi:10.1007/s10864-010-9099-0
- Spencer, T. D., & Higbee, T. S. (2012). Using transfer of stimulus control technology to promote generalization and spontaneity of language. *Focus on Autism and Other Developmental Disabilities*, 27, 225–236. doi:10.1177/1088357612460274
- Sprague, J. R., & Horner, R. H. (1984). The effects of single instance, multiple instance, and general case training on generalized vending machine use by moderately and severely handicapped students. *Journal of Applied Behavior Analysis*, 17, 273–278. doi:10.1901/jaba.1984.17-273

Stokes, T. F., & Baer, D. M. (1977). An implicit technology of generalization. *Journal of Applied Behavior Analysis*, 10, 349–367. doi:10.1901/jaba.1977.10-349

Stokes, T. F., & Osnes, P. G. (1989). An operant pursuit of generalization. *Behavior Therapy*, 20, 337–355. doi:10.1016/S0005-7894(89)80054-1

Swan, A. J., Carper, M. M., & Kendall, P. C. (2015). In pursuit of generalization: An

updated review. *Behavior Therapy*, 47, 733–746. doi:10.1016/j.beth.2015.11.006

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C H A P T E R 2 2

NATURALISTIC TEACHING

Keith D. Allen and Stacy B. Shaw

OVERVIEW: WHAT WE KNOW

Early behavioral treatments for individuals with autism used basic principles of learning to create highly effective teaching interactions. Their success was largely a result of highly structured training procedures, which were developed specifically to maximize learning potential. Distractions were minimized, prompts were delivered in a precise sequence, practices were massed, and consequences were delivered in a reliable, immediate, and frequent manner. These tightly controlled training procedures were also known as "discrete trial training" and have been found to be quite effective at teaching new and complex skills. Indeed, these intensive teaching procedures led to impressive gains in language, social, and adaptive skills for many individuals with autism.

Unfortunately, these highly structured approaches to learning were not without problems. Of significant concern was the fact that many of the newly acquired skills would be exhibited in the highly structured learning environment but not in the less structured natural environment. In other words, the skills learned in controlled training settings did not automatically transfer, or generalize, to other important settings (e.g., regular classroom, playground), other important people (e.g., peers, parents), or other important tasks (e.g., play, daily living tasks). For example, a child who had learned to say "truck" when presented with a picture of a truck in a tightly controlled treatment setting would not necessarily say "truck" when he wanted access to his toy truck at home. Gradually it became clear that in order for important skills to be used in the natural settings of everyday life, training would need to be systematically planned and implemented in a way that would promote generalization.

Over recent years, "naturalistic" teaching has emerged as a way to address this problem with generalization. Naturalistic teaching promotes generalization of skills to the natural or everyday settings where those skills are required. Recent reviews of the expanding literature on naturalistic teaching have found that the research support for this approach is extensive. (See Allen, K. D., & Cowen, R. J., (2008) for a review.) Naturalistic procedures have been found to enhance the generalization of language, social, and play skills. In direct comparisons with the more structured approaches, naturalistic procedures have, in many situations, been found to produce better

generalization of important skills to the natural environment. In addition, because naturalistic procedures occur within the context of everyday life and activities, they can help make learning more fun and in many cases increase a child's willingness to participate in teaching interactions. As a result, parents and practitioners can feel confident in using naturalistic procedures as part of an evidence-based approach to educating individuals with autism.

GETTING STARTED: PRINCIPLES AND APPLICATIONS

To understand the basic principles of naturalistic teaching, a distinction must be made between two behavioral processes: discrimination and generalization. When an individual exhibits the correct or desired response under certain stimulus conditions and not others, the individual is said to have discriminated. Thus, an individual who selects a truck when asked to select the truck from a choice of options such as truck, cup, and shoe has discriminated between different choices.

Generalization occurs when that same correct or desired response occurs under similar but untrained stimulus conditions. Thus, the individual who can select the truck when the truck looks different (e.g., different size, color, or function) from that which was used during training is said to have demonstrated generalization. In addition, an individual who can make the correct response when the teacher or the setting is different is also said to have demonstrated generalization.

Discrimination training can be a critical part of learning because sometimes it is very important that an individual exhibit certain responses under very specific conditions and only those conditions. For example, it is important that individuals learn "stranger danger" and to not get in the car with anyone other than a parent or guardian. This requires that "coming when called" does not generalize to every adult who is calling. Likewise, individuals must learn that not all four-legged animals are dogs, not all liquids are juice, not all coins are quarters, and not all buses will arrive downtown.

Discrimination is problematic, however, when it leads to an individual being unable to perform certain responses under many different yet important conditions. For example, an individual must learn to ask for help not just from the teacher but also from a boss, a parent, or, at times, a peer. Likewise, an individual must learn to identify a jacket on the coat rack as well as in a picture, to count coins at the store as well as at school, and to recite his or her address to a cab driver as well as a parent. When individuals do not generalize important language, social, and play skills from the training environment to everyday life, difficulties in adaptive functioning may result.

GUIDELINES FOR IMPLEMENTATION

When generalization is the goal, naturalistic teaching can be an important solution. However, there is no single or gold standard naturalistic teaching procedure to guide implementation. Various approaches can be found under a number of different names, including incidental teaching, natural language paradigm, behavioral chain interruption, script fading, and milieu teaching. Each procedure emphasizes slightly different ways to approach learning and generalization.

What follows are guidelines, derived from these different procedures, that represent an integration of the most distinguishing characteristics of naturalistic teaching.

TEACH RESPONSES THAT WILL LIKELY CREATE CONTACT WITH NATURAL REINFORCERS

Generalization is enhanced when an individual is taught a response for which the natural consequence is a reinforcer—that is, a consequence that strengthens responding. Natural consequences are those that would occur naturally, without needing to be programmed by the teacher. For example, a natural consequence to a request for a “truck” is to hand the individual a truck. Thus, the truck naturally reinforces the individual for saying “truck.” The use of praise, hugs, or even candy as a consequence would not be a natural consequence of a request for a truck, and these arbitrary consequences would require specific programming by the teacher. Furthermore, responding in this latter case may be limited to conditions where the teacher has arranged for these types of arbitrary reinforcement. Because naturalistic teaching targets responses that will come into contact with naturally available reinforcers across conditions, generalization is enhanced.

Here is an example. Zach’s teacher, Mrs. Cole, has observed that Zach loves playing with cars and trucks during free play. Mrs. Cole has decided to target Zach’s expressive language by using natural reinforcers that are available. When Zach shows interest, he will be required to request to play with the cars and trucks and to elaborate by describing features of the ones he wants first (e.g., “Yes, Zach, you may play with the cars and trucks. Which color cars and trucks would you like?” or “How many would you like?”). She will hand him a few and once he is engaged, she will impose the same expectations before Zach can access more or different kinds of cars or trucks (i.e., she will require request + elaboration).

ALLOW AN INDIVIDUAL’S INTERESTS TO DIRECT AND PACE TEACHING

Given the emphasis on natural reinforcers, there will be times when teaching cannot occur because an individual, for whatever reason (e.g., illness, fatigue, satiation), is not interested in the current activities or materials. Under highly structured teaching conditions, a teacher might try to overcome this disinterest by introducing arbitrary reinforcers—for example, the use of candy to reinforce turn taking with the trucks, praise to reinforce requesting, access to preferred computer games to reinforce following directions. However, because naturalistic teaching relies on natural reinforcers, teaching is not forced when interest has waned.

Here are several examples. During free time, Mrs. Cole has planned to work with Zach on appropriate sharing and play sequences with cars and trucks as well as on expressive language requests and elaborations, but today Zach has followed several children to the kitchen area, where he seems interested in setting the table with plastic dishes, cups, and food. Mrs. Cole realizes that at this moment naturalistic teaching associated with car and truck play is not available. Instead, she follows Zach to the kitchen, takes several plates and pieces of food, holds them within view, and waits for Zach to request them. When he does, she asks him to elaborate the colors, sizes, and categories (e.g., fruits or vegetables) that he wants and then hands them to him. Later, Zach loses interest in the kitchen and begins pacing the room. Although she had hoped to

work with him on turn taking, she recognizes that because his interest is currently low, naturalistic teaching is not appropriate at this time.

EMBED TEACHING WITHIN THE ACTIVITIES OF EVERYDAY LIFE

Generalization is enhanced when training conditions are diverse and loosely structured. The goal of this approach is for the individual to be unable to tell where or when training is occurring so that trained responses occur everywhere and with everyone. By embedding teaching within the activities of everyday life, diversity is increased by the inclusion of multiple “teachers” (e.g., paraeducators, parents, peers) across multiple settings (e.g., on the playground, in the hallways, in the backyard at home, during a bath, at the mall) and at various times throughout the day.

Here are some examples. Mrs. Cole has observed that Zach loves playing on swings and slides on the playground. Mrs. Cole has decided to target Zach’s expressive language by using natural reinforcers that are available during recess. Zach will be required to request to go outside and to elaborate by describing which activity he wants to engage in first (e.g., “Yes, Zach, you may go outside. Where do you want to play first?”). Mrs. Cole has recruited Zach’s parents and his daycare providers to do the same. Mrs. Cole has also recruited several peers to participate as “peer tutors” to assist with naturalistic teaching. When Zach is observed to be engaged with cars and trucks, she prompts a peer to arrive with additional highly preferred cars and trucks. The peers then share the additional cars with Zach contingent upon his appropriate request. At home, Zach’s parents also make cars, puzzles, and computers accessible only after appropriate requests and elaborations by Zach. The teacher provides a list of the types of elaborations they are working on at school (e.g., “Sure, you can play with some cars for a while. Which color cars would you like?”). In addition, Zach’s parents model pushing the cars, making car sounds, and turn taking, encouraging Zach to imitate. They also arrange play dates with a selected peer who brings preferred snacks and puzzles that Zach can access by asking the peer what he has in his backpack. The peer also models appropriate play sequences and turn taking with toys. Back at school, Mrs. Cole occasionally sends Zach on an errand to the office. She has recruited the office secretary to dispense information about the errand only after Zach queries for that information. The secretary also has a collection of treats Zach likes. Mrs. Cole has recruited the secretary to keep the treats in one of three tins on her desk. Zach must query for information (i.e., “Where is it?”), request access, and elaborate about what color, size, or number of treats he would like.

INCORPORATE PROMPTS THAT CAN BE TRANSPORTED TO NEW SITUATIONS

Prompts are most often thought of as teaching methods that help promote learning. However, generalization can also be enhanced by including prompts within one training situation that can easily be transported to new situations.

For example, individuals with autism often do not initiate social-communicative interactions in any setting and, as a result, prompting is often required. The mand-model procedure allows a teacher to initiate an interaction by telling the child what is required and then modeling the appropriate response. For example, Mrs. Cole might have some nifty cars, but Zach only looks or grabs and does not request without a prompt. Mrs. Cole might then say, “What do you want?”

and immediately model, "Say, 'I want a car.'" Mrs. Cole might also introduce a time-delay procedure where she gradually increases the time between her request and her model, giving Zach a chance to respond but helping him before he gets frustrated or loses interest. Mrs. Cole could also use a script-fading procedure in which visual and/or auditory cues or prompts are provided in the natural setting. For example, Mrs. Cole wants to teach Zach how to approach a group at recess and request sharing of access to a swing. She decides to teach him to follow a written script of an appropriate interaction. Because this is a new skill for Zach, Mrs. Cole initially prompts and reinforces script following in a tightly controlled, less naturalistic setting. Eventually, as Zach acquires the skill, he might take the script with him to recess (e.g., on a small laminated card in his back pocket), where another teacher prompts Zach to look at and follow the script, as needed.

ADAPTATIONS AND MODIFICATIONS

RESTRICT ACCESS

One of the common obstacles to naturalistic teaching is that too often opportunities for naturalistic teaching are lost because individuals have rather unlimited or only slightly restricted access to the things that interest them most. However, with some increased restrictions and a touch of creativity, many additional naturalistic teaching opportunities can be created.

Here are some examples. Mrs. Cole has taken the time to arrange the classroom so that some of Zach's most preferred activities and objects are visible but not easily accessed. His favorite cars and puzzles are on the top of a cabinet that can be accessed only by adults. When he arrives each day his dessert and drink are removed from his lunch and he is required to initiate additional requests to access them. His coat and boots are also moved and after a request to go outside, he is prompted to ask a classroom aide (or sometimes peer tutor) for their location. At home, computer games he enjoys are locked in a cabinet and the best DVD is always in a novel location, requiring queries and social exchanges with parents and sometimes siblings who are recruited to assist with naturalistic teaching.

INTERRUPT CHAINS

Although much of naturalistic teaching occurs at the beginning of potential interactions, when motivation is high, there are times when naturalistic teaching can occur in the middle of an activity sequence. *Behavior chain interruption* involves an interruption of the chain or flow of purposeful behavior in the midst of a familiar routine. For example, Zach's parents typically require Zach to request to play a computer game, but they have also purchased a timer that attaches to the computer. The timer allows them to limit his time on the computer. When the time elapses, a 60-second countdown warns Zach of impending shutdown. Zach's play is then interrupted and he must immediately find his parents and initiate a request for more time.

Zach's parents have also taught him to make his own peanut-butter-and-jelly sandwich. They did this initially in a more controlled, discrete-trial type of procedure. Once mastered, his parents intentionally arrange for him to periodically discover that there is only one piece of bread or a very small amount of jelly. Solving the problem requires interrupting his activities and asking his parents for help.

KEYS TO REMEMBER

1. Naturalistic teaching promotes generalization of skills to settings where those skills are required.
2. To find good naturalistic teaching opportunities, follow the individual's interests and activities.
3. Learning does not need to occur just in the classroom or in tightly controlled, distraction-free environments. Look for opportunities to teach within the context of everyday events with everyday people.
4. Not all skills can be learned through naturalistic teaching. Some new or complex skills may be better learned in a more controlled setting before taking that skill into the natural setting.

SUGGESTED READINGS

- Allen, K. D., & Cowen, R.J. (2008). Using naturalistic procedures to teach children with autism. In J. Luiselli, D. Russo, W. Christian, & S. Wilczynski (Eds.), *Effective Practices for Children with Autism: Educational and Behavior Support Interventions that Work*. New York: Oxford University Press.
- Alpert, C., & Kaiser, A. (1992). Training parents as milieu language teachers. *Journal of Early Intervention*, 16, 31–52.
- Kohler, F. W., Strain, P. S., Hoyson, M., & Jamieson, B. (1997). Merging naturalistic teaching and peer-based strategies to address the IEP objectives of preschoolers with autism: An examination of structural and child behavior outcomes. *Focus on Autism and Other Developmental Disabilities*, 12, 196–206.
- Stokes, T. F., & Osnes, P.G. (1989). The operant pursuit of generalization. *Behavior Therapy*, 20, 337–355.
- Sundberg, M., & Partington, J. (1999). The need for both discrete trial and natural environment language training for children with autism. In P. Ghezzi, W. Williams, & J. Carr (Eds.), *Autism: Behavior Analytic Perspectives* (pp. 139–156). Reno, NV: Context Press.

PROMPTING

CHARACTERISTICS OVERVIEW CHART

Verbal Skills	Grade Levels	Cognitive Level	Areas Addressed
☒ Nonverbal	☒ PK	☒ Classic	☒ (Pre)Academic/Cognitive/Academic
☒ Mixed	☒ Elementary	☒ High Functioning	☒ Adaptive Behavior/ Daily Living
☒ Verbal	☒ Middle/High		☒ Behavior ☒ Communication/Speech ☒ Social/Emotional

BRIEF INTRODUCTION

Prompting is extensively used in behavior shaping and skill acquisition. It provides learners with assistance to increase the probability that a desired behavior will occur. Successful performance of a desired behavior elicits positive reinforcement, therefore reinforcing learning. Prompts vary from most to least intrusive. Prompting should be faded to avoid prompt dependency.

DESCRIPTION

Prompting is a means to induce an individual with added stimuli (prompts) to perform a desired behavior. A prompt is like a cue or support to encourage a desired behavior that otherwise does not occur. In other words, a prompt is an antecedent that is provided when an ordinary antecedent is ineffective. Prompts are often categorized into a hierarchy from most intrusive to least intrusive. Types of prompts (from most intrusive to least intrusive), their descriptions, and examples are as follows:

- *Full physical assistance.* The teacher uses “hand-over-hand” support to aid the child in completing a task (e.g., when teaching the child to pick up a cup, the teacher takes the child’s hand and guides him to pick it up).

- *Partial physical assistance.* The teacher provides partial physical assistance to help the child complete a task (e.g., when teaching the child to pick up the cup, the teacher guides the child's hand to the cup by tapping his elbow).
- *Full model.* The teacher models the desired behavior (e.g., when teaching the child how to clap, the teacher claps while telling the child to clap).
- *Partial model.* The teacher models only part of the desired behavior (e.g., when teaching the child how to clap, the teacher puts his hands in front of himself, but does not actually clap).
- *Full verbal prompts.* The teacher verbally models the desired behavior (e.g., when teaching the child to expressively label "car," the teacher asks, "What is it? Say car.>").
- *Partial verbal model.* The teacher verbally models only part of the desired behavior (e.g., when teaching the child to expressively label "car," the teacher asks, "What is it? Say c____").
- *Gestural prompt.* The teacher utilizes a physical gesture to encourage the desired behavior (e.g., when teaching the function of an object, the teacher says, "What do you drink with?" while holding his hand to his mouth shaping it like a cup).
- *Positional prompt.* The teacher places the target item in a location that is closer to the child (e.g., when teaching the child to label "toy," the teacher places the toy closest to the child).
- *Time-delay or prompt-delay techniques* (Walker, 2008). This instructional procedure is proven to be effective, especially for children with AU. When teaching a novel task, time delay is used to transfer the stimulus control from a controlling prompt to a natural prompt by placing varying amounts of time between a controlling prompt and a natural prompt. Given different lengths of time delay, time delay strategies are categorized into constant time delay (CTD) and progressive time delay (PTD). CTD indicates that there is a standard time delay whereas PTD has a graduated delay. The procedures of time delay strategy begin with a zero-second (0-s) delay trial, meaning the controlling prompt is presented with task instruction at the same time without any delay in between. Gradually, to fade the prompt, time delay is increased between the natural prompt (task direction) and the controlling prompt.

Not all prompts in the hierarchy need to be used when teaching a skill. Prompts should be chosen based on which ones are most effective for a particular child. Prompts should be faded systematically and as quickly as possible to avoid prompt dependency. Overall, the goal of using prompts is to help the child independently perform the desired behavior.

STEPS

1. *Identify the least intrusive prompt.* Choose a prompt that is necessary for a correct response to occur.
2. *Give differential reinforcement.* After a correct response, give appropriate reinforcement that is equivalent to the level of performance independency.
3. *Fade prompt.* After the child masters a skill, gradually move prompt away or replace with least intrusive prompt.

BRIEF EXAMPLE

Josh was learning to match objects with names. The teacher started with verbal and gestural prompts. She said, “It is time for drawing, and we need some crayons to draw with.” She looked at the crayons on the table and pointed at them, saying, “These are crayons.” Then she pointed at the crayons and asked Josh, “What are these?” After Josh responded “Crayons,” the teacher nodded her head and said, “These are crayons! Good job, Josh!”

To fade prompts, the teacher gradually moved from verbal and gestural prompts to only positional prompts. For example, the next time the teacher placed the crayons on the table near Josh, she said, “We need some crayons to draw with. Josh, show me the crayons.” Josh pointed to the crayons correctly and received a point on his token board.

SUMMARY

Prompting is used to increase desired behaviors and skill acquisition. When delivering prompts, the instructor should be mindful of the student's learning level and fading considerations.

Reinforcement should occur after a correct response and should be contingent upon the level of performance independence desired.

RESEARCH TABLE

Number of Studies	Ages (year)	Sample Size	Area(s) Addressed	Outcome
48*	3 to adult	139	Academics, art, communication, imitative skill, generalization, social skills, daily living skills, leisure choice, Internet skills, spontaneous response, disruptive behaviors, reading comprehension, safety, writing, empathy, observation, on task, identifying numbers	+

*Note: Includes an integrated review of literature by Walker (2008).

STUDIES CITED IN RESEARCH TABLE

1. Batchelder, A., McLaughlin, T. F., Weber, K. P., Derby, K. M., & Gow, T. (2009). The effects of hand-over-hand and a dot-to-dot tracing procedure on teaching an autistics student to write his name. *Journal of Developmental and Physical Disabilities, 21*, 131-138.
The purpose of the present study was to evaluate the effects of hand-overhand and dot-to-dot tracing procedure to teach a 14-year-old student with classic autism to write his first name. A multiple baseline and reversal design was used to evaluate hand-over-hand and dot-to-dot tracing. The results indicated the participant increased his ability to write the letters in his first name from 60% to 100%.
2. Karsten, A. M., & Carr, J. E. (2009). The effects of differential reinforcement of unprompted responded on the skill acquisition of children with autism. *Journal of Applied Behavior Analysis, 42*(2), 327-334.
The purpose of the investigation was to compare the delivery of high-quality reinforcers exclusively following unprompted responses (differential reinforcement) with the delivery of high-quality reinforcers following both prompted and unprompted responses (nondifferential reinforcement) on the skill acquisition of two children with autism (ages 3 and 5). Results indicated that both were effective teaching procedures, although the differential reinforcement procedure was more reliable in producing skill acquisition.

3. Schrandt, J., Townsend, D. B., & Poulson, C. L. (2009). Teaching empathy skills to children with autism. *Journal of Applied Behavior Analysis, 42*(1), 17-32.
The purpose of this study was to teach empathetic responding to four children with autism (ages 4 to 8). Instructors presented vignettes with dolls and puppets demonstrating various types of affect and used prompt delay, modeling, manual prompts, and reinforcement to teach participants empathy responses. Increases in empathetic responding occurred systematically with the introduction of treatment across all participants. Furthermore, responding generalized from training to nontraining stimuli for all participants. Generalization occurred from dolls and puppets to actual people in a nontraining setting for two participants.
4. Ledford, J. R., Gast, D. L., Luscre, D., & Ayres, K. M. (2008). Observational and incidental learning by children with autism during small group instruction. *Journal of Autism and Developmental Disorders, 38*, 86-103.
This study evaluated the acquisition of incidental and observational information presented to six children with autism (ages 5 to 8 years old) using a time delay procedure. Students learned observational and incidental information during small group instruction.
5. Liber, D. B., Frea, W. D., & Symon, J. B. G. (2008). Using time delay to improve social play skills with peers for children with autism. *Journal of Autism and Developmental Disorders, 38*, 312-323.
The study involved three boys with autism who were taught play activities that combined a play sequence with requesting peer assistance using a graduated time-delay procedure. Results showed an increase in pretend play by one of the participants. Two also demonstrated a generalization of the skills learned through the time-delay procedure.
6. Walker, B. (2008). Constant and progressive time delay procedures for teaching children with autism: A literature review. *Journal of Autism and Developmental Disorders, 38*, 261-275.
This review of 22 studies with 53 children with autism on time delay revealed that constant and progressive time-delay procedures were effective, with the latter resulting in a lower error rate and fewer procedural errors.
7. Jerome, J., Frantino, E. P., & Sturmey, P. (2007). The effects of errorless learning and backward chaining on the acquisition of Internet skills in adults with developmental disabilities. *Journal of Applied Behavior Analysis, 40*, 185-189.
Three adults with autism and mental retardation were taught to access specific Internet sites using backward chaining and most-to-least intrusive prompting. Results indicated that the number of independent steps completed in the task analysis increased following training.
8. Jones, E. A., Feeley, K. M., & Takacs, J. (2007). Teaching spontaneous responses to young children with autism. *Journal of Applied Behavior Analysis, 40*, 565-570.
The study evaluated the effectiveness of intensive intervention in establishing spontaneous verbal responses to two young children with autism. The intervention involved discrete-trial

instruction, specific prompts, and error corrections. Data showed positive outcomes of the intervention.

9. Murzynski, N. T., & Bourret, J. C. (2007). Combining video modeling and least-to-most prompting for establishing response chains. *Behavioral Interventions*, 22, 147-152.
Two boys with autism participated in the study, which compared video modeling combined with least-to-most intrusive prompting to least-to-most intrusive prompting alone in teaching daily living skills in the form of response chains. The results showed that the participants acquired skills taught with video modeling plus least-to-most prompting in fewer trials and with fewer prompts than skills taught with least-to-most prompting alone.
10. Martin, C. A., Drasgow, E., Halle, J. W., & Brucker, J. M. (2005). Teaching a child with autism and severe language delays to reject: Direct and indirect effects of functional communication training. *Educational Psychology*, 25, 287-304.
The study examined the effects of direct functional communication training, involving prompting, differential reinforcement, and error correction procedures, in teaching a 10-year-old child with autism to reject items by touching an icon. Results indicated that the training was successful at replacing pushing away with touching an icon to reject items, but it had variable effects on the other behaviors serving a rejecting function.
11. Akmanoglu, N., & Batu, S. (2004). Teaching pointing to numerals to individuals with autism using simultaneous prompting. *Education and Training in Developmental Disabilities*, 39, 326-336.
The purpose of this study was to examine effectiveness of simultaneous prompting in teaching pointing to numerals to individuals with autism. Three individuals with autism (ages 6, 12, and 17) were taught pointing to numerals, which were orally named by the teacher using simultaneous prompting. Results revealed that simultaneous prompting was effective in teaching pointing to numerals. Participants generalized this skill to pointing to the numerals on a calendar page.
12. Lerman, D. C., Vorndran, C., Addison, L., & Kuhn, S.A.C. (2004). A rapid assessment of skills in young children with autism. *Journal of Applied Behavior Analysis*, 37, 11-26.
The relative efficacy of two assessment packages, one contained several reinforcement procedures while the other contained several potentially effective prompts, was evaluated across two to three skills for six children with autism. Results suggested that the methodology was useful for matching targeted skills to appropriate interventions.
13. Morrison, R., Sainato, D., BenChaaban, D., & Endo, S. (2002). Increasing play skills of children with autism using activity schedules and correspondence training. *Journal of Early Intervention*, 25, 58-72.
The purpose of this study was to investigate a strategy to improve the independent performance of four preschoolers with autism during playtime in an inclusive setting. Results of the study indicated that all four participants' on-task and play correspondence behavior increased, while experimenter prompts gradually decreased.

14. Bryan, L.C., & Gast, D.L. (2000). Teaching on-task and on-schedule behaviors to high-functioning children with autism via picture activity schedules. *Journal of Autism and Developmental Disorders*, 30, 553-567.

The purpose of this investigation was to evaluate the effectiveness of using graduated guidance and visual activity schedules to teach four elementary students with autism on-task and on-schedule behavior. Student performance (a) increased when prompting was used; (b) performance was maintained when the picture activity book was available; (c) it dropped when the picture activity book was not available; and (c) student performance generalized to novel activities.

15. Massey, G., & Wheeler, J. (2000). Acquisition and generalization of activity schedules and their effects on task engagement in a young child with autism in an inclusive preschool classroom. *Education and Training in Mental Retardation and Developmental Disabilities*, 35, 326-335.

Efficacy of individualized activity schedules with a 4-year old child with autism was examined. Graduated physical guidance and a system of most-to-least prompts were used during skill acquisition stages with teacher proximity and level of prompts being eventually faded. The child successfully acquired the skills necessary to independently follow activity schedules and generalized these skills to other settings with minimal training.

16. Buffington, D. M., Krantz, P. J., McClannahan, L. E., & Poulson, C. L. (1998). Procedures for teaching appropriate gestural communication skills to children with autism. *Journal of Autism and Developmental Disorders*, 28, 535-545.

Four children with autism were taught to use gestures in combination with oral communication. Although none of the participants displayed appropriate gestural and verbal responses during baseline, they all acquired this skill with the systematic implementation of modeling, prompting, and reinforcement. The skills learned were generalized.

17. Freeman, K. A., & Piazza, C. C. (1998). Combining stimulus fading, reinforcement, and extinction to treat food refusal. *Journal of Applied Behavior Analysis*, 31, 691-694.

The study combined the procedures of fading, reinforcement, and escape extinction using guided compliance to increase food consumption while maintaining low rates of disruptive behavior at mealtime for a 6-year-old girl with autism. Results indicated that intake increased and compliance with prompting procedures remained relatively stable despite a requirement to increase consumption.

18. Strain, P. S., Kohler, F. W., Storey, K., & Danko, C. D. (1994). Teaching preschoolers with autism to self-monitor their social interactions: An analysis of results in home and school settings. *Journal of Emotional and Behavioral Disorders*, 2, 78-88.

Three preschool boys with autism participated in the study, which examined the effects of a self-monitoring intervention. Three primary results were obtained. First, the self-monitoring package increased each participant's interactions with his peers and/or siblings. Second, the school and home procedures produced comparable impacts on some dimensions of children's social

behaviors, but other outcomes were affected differentially. Finally, both adult prompts and reinforcement were successfully reduced or faded within the school and home intervention settings.

19. Carr, E. G., & Darcy, M. (1990). Setting generality of peer modeling in children with autism. *Journal of Autism and Developmental Disorders*, 20, 45-59.
Four preschoolers with autism participated in an intervention in which a typical peer demonstrated and physically prompted a variety of action and object manipulations that defined the activity. Following training, all four preschoolers generalized their imitative skill to a new setting involving new actions and object manipulations.
20. Carr, E. G., & Kemp, D. C. (1989). Functional equivalence of autistic leading and pointing: Analysis and treatment. *Journal of Autism and Developmental Disorders*, 19, 561-578.
A less desirable form of requesting, leading was treated by strengthening a more desirable form of requesting, pointing. The study was conducted with four children (aged 3 and 5 yrs) with autism. Intervention included verbal and physical prompting of the pointing response as well as tangible reinforcement for child-initiated instances of that response. Verbal requesting was also taught to accompany the pointing. Following intervention, generalization was observed.
21. Godby, S., Gast, D. L., & Wolery, M. (1987). A comparison of time delay and system of least prompts in teaching object identification. *Research in Developmental Disabilities*, 8, 283-306.
Two prompting procedures, progressive time delay and a system of least prompts, were used to teach three students (aged 8, 11, and 16 yrs) to identify functional objects. Both prompting procedures were effective in establishing correct responding at criterion levels, but the time delay procedure was more efficient than the system of least prompts.
22. MacArthur, J., Ballard, K. D., & Artinian, M. (1985). Teaching independent eating to a developmentally handicapped child showing chronic food refusal and disruption at mealtimes. *Australia & New Zealand Journal of Developmental Disabilities*, 12, 203-210.
The study describes an intervention to establish independent eating behaviors in a 3-year-old boy with autistic-like behaviors. The first phase of the intervention, using backward chaining with prompting and fading of prompts, reduced screaming, food refusal, and related disruptive mealtime behaviors and established appropriate eating responses. The second phase of the intervention successfully taught the mother to implement the teaching strategies, and established independent mealtime behaviors at home. In addition, follow-up showed maintenance of independent eating.
23. Hawkins, A. H. (1982). Influencing leisure choices of autistic-like children. *Journal of Autism and Developmental Disorders*, 12, 359-366.
In the study, a leisure education program was implemented with three boys with autistic-like behaviors. Three alternating conditions (prompt and praise condition, praise anything condition, and token economy condition) were used to encourage participation in low-

interest recreational activities. Results showed that the prompt and praise condition was superior to the other two conditions in encouraging participation in low-interest recreational activities.

24. Schepis, M. M. et al. (1982). A program for increasing manual signing by autistic and profoundly retarded youth within the daily environment. *Journal of Applied Behavior Analysis*, 15, 363-379. The study investigated a modified incidental teaching strategy program that included re-arranging the physical environment to prompt signing, altering routine staff-resident interactions to prompt, manually guiding and/or reinforcing signing, and conducting mini-training sessions (5 profoundly retarded and 4 autistic youth). Results indicated significant increases in signing for all participants. The increases were maintained during follow-up. Social validity of the program was also documented.
25. Schreibman, L., & Charlop, M. H. (1981). S+ versus S- fading in prompting procedures with autistic children. *Journal of Experimental Child Psychology*, 31, 508-520. The study investigated the relative effectiveness of two prompt-fading procedures for teaching difficult visual discriminations to eight children with autism. Both prompt procedures involved within-stimulus fading whereby manipulation occurred on the relevant component of the discrimination. One procedure involved fading first along the positive stimulus (S+), while holding the negative stimulus (S-) constant. The other procedure involved fading first along the S-, while holding the S+ constant. Results indicate that for all but one child, the discriminations were acquired significantly faster with fewer errors when the positive stimulus was faded first.
26. Rincover, A. (1978). Variables affecting stimulus fading and discriminative responding in psychotic children. *Journal of Abnormal Psychology*, 87, 541-553. Two experiments were documented. The first investigated stimulus variables in prompt fading that might reduce the attentional requirements for discrimination learning of eight children with autism. Two variables were assessed, distinctive vs. nondistinctive feature fading. Significant main effects were found for both variables due to the success of the within-stimulus and the distinctive feature conditions. The combination of within-stimulus and distinctive feature fading was the most effective procedure. The second experiment was conducted with the same children to assess whether they were still responding only to the pretrained feature after fading. Results showed that discriminative responding was maintained when the pretrained feature was made irrelevant, demonstrating that children attended to multiple features of a positive stimulus, but it was disrupted when the whole letter containing the pretrained feature was made irrelevant, showing that children still learned a restricted portion of the positive reinforcement.
27. Rosenbaum, M. S., & Breiling, J. (1976). The development and functional control of reading-comprehension behavior. *Journal of Applied Behavior Analysis*, 9, 323-333. In the study, a 12-year-old girl was taught reading comprehension using verbal prompts, modeling, and physical guidance. The girl was rewarded for correct behaviors in response to the training item. Results revealed that (a) following acquisition, performance was maintained at

near a 100% level when candy, praise, attention, and training were removed; (b) absence of other persons was correlated with a marked decrease in performance, whereas their presence was associated with performance at near 100%; and (c) performance generalized to probes and across experimenters. Rewards, which may have been reinforcing during acquisition, did not appear necessary to maintain later performance. Instead, the presence of others had control over maintenance performance.

REFERENCES

- Akmanoglu, N., & Batu, S. (2004). Teaching pointing to numerals to individuals with autism using simultaneous prompting. *Education and Training in Developmental Disabilities, 39*, 326-336.
- Batchelder, A., McLaughlin, T. F., Weber, K. P., Derby, K. M., & Gow, T. (2009). The effects of hand-over-hand and a dot-to-dot tracing procedure on teaching an autistics student to write his name. *Journal of Developmental and Physical Disabilities, 21*, 131-138.
- Bryan, L.C., & Gast, D.L. (2000). Teaching on-task and on-schedule behaviors to high-functioning children with autism via picture activity schedules. *Journal of Autism and Developmental Disorders, 30*, 553-567.
- Buffington, D. M., Krantz, P. J., McClannahan, L. E., & Poulson, C. L. (1998). Procedures for teaching appropriate gestural communication skills to children with autism. *Journal of Autism and Developmental Disorders, 28*, 535-545.
- Carr, E. G., & Darcy, M. (1990). Setting generality of peer modeling in children with autism. *Journal of Autism and Developmental Disorders, 20*, 45-59.
- Carr, E. G., & Kemp, D. C. (1989). Functional equivalence of autistic leading and pointing: Analysis and treatment. *Journal of Autism and Developmental Disorders, 19*, 561-578.
- Freeman, K. A., & Piazza, C. C. (1998). Combining stimulus fading, reinforcement, and extinction to treat food refusal. *Journal of Applied Behavior Analysis, 31*, 691-694.
- Godby, S., Gast, D. L., & Wolery, M. (1987). A comparison of time delay and system of least prompts in teaching object identification. *Research in Developmental Disabilities, 8*, 283-306.
- Hawkins, A. H. (1982). Influencing leisure choices of autistic-like children. *Journal of Autism and Developmental Disorders, 12*, 359-366.
- Jerome, J., Frantino, E. P., & Sturmey, P. (2007). The effects of errorless learning and backward chaining on the acquisition of Internet skills in adults with developmental disabilities. *Journal of Applied Behavior Analysis, 40*, 185-189.

- Jones, E. A., Feeley, K. M., & Takacs, J. (2007). Teaching spontaneous responses to young children with autism. *Journal of Applied Behavior Analysis*, 40, 565-570.
- Karsten, A. M., & Carr, J. E. (2009). The effects of differential reinforcement of unprompted responded on the skill acquisition of children with autism. *Journal of Applied Behavior Analysis*, 42(2), 327-334.
- Ledford, J. R., Gast, D. L., Luscre, D., & Ayres, K. M. (2008). Observational and incidental learning by children with autism during small group instruction. *Journal of Autism and Developmental Disorders*, 38, 86-103.
- Lerman, D. C., Vorndran, C., Addison, L., & Kuhn, S. A. C. (2004). A rapid assessment of skills in young children with autism. *Journal of Applied Behavior Analysis*, 37, 11-26.
- Liber, D. B., Frea, W. D., & Symon, J. B. G. (2008). Using time delay to improve social play skills with peers for children with autism. *Journal of Autism and Developmental Disorders*, 38, 312-323.
- MacArthur, J., Ballard, K. D., & Artinian, M. (1985). Teaching independent eating to a developmentally handicapped child showing chronic food refusal and disruption at mealtimes. *Australia & New Zealand Journal of Developmental Disabilities*, 12, 203-210.
- Martin, C. A., Drasgow, E., Halle, J. W., & Brucker, J. M. (2005). Teaching a child with autism and severe language delays to reject: Direct and indirect effects of functional communication training. *Educational Psychology*, 25, 287-304.
- Massey, G., & Wheeler, J. (2000). Acquisition and generalization of activity schedules and their effects on task engagement in a young child with autism in an inclusive preschool classroom. *Education and Training in Mental Retardation and Developmental Disabilities*, 35, 326-335.
- Morrison, R., Sainato, D., BenChaaban, D., & Endo, S. (2002). Increasing play skills of children with autism using activity schedules and correspondence training. *Journal of Early Intervention*, 25, 58-72.
- Murzynski, N. T., & Bourret, J. C. (2007). Combining video modeling and least-to-most prompting for establishing response chains. *Behavioral Interventions*, 22, 147-152.
- Rincover, A. (1978). Variables affecting stimulus fading and discriminative responding in psychotic children. *Journal of Abnormal Psychology*, 87, 541-553.
- Rosenbaum, M. S., & Breiling, J. (1976). The development and functional control of reading-comprehension behavior. *Journal of Applied Behavior Analysis*, 9, 323-333.

- Schepis, M. M. et al. (1982). A program for increasing manual signing by autistic and profoundly retarded youth within the daily environment. *Journal of Applied Behavior Analysis*, 15, 363-379.
- Schreibman, L., & Charlop, M. H. (1981). S+ versus S- fading in prompting procedures with autistic children. *Journal of Experimental Child Psychology*, 31, 508-520.
- Schrandt, J., Townsend, D. B., & Poulson, C. L. (2009). Teaching empathy skills to children with autism. *Journal of Applied Behavior Analysis*, 42(1), 17-32.
- Strain, P. S., Kohler, F. W., Storey, K., & Danko, C. D. (1994). Teaching preschoolers with autism to self-monitor their social interactions: An analysis of results in home and school settings. *Journal of Emotional and Behavioral Disorders*, 2, 78-88.
- Walker, B. (2008). Constant and progressive time delay procedures for teaching children with autism: A literature review. *Journal of Autism and Developmental Disorders*, 38, 261-275.

RESOURCES AND MATERIALS

- Antecedent Prompting for Children with Autism: <http://www.teachervision.fen.com/autism/teaching-methods/8201.html>
This Teachervision article presents information on using prompting in the classroom.
- Prompting. Evidence-Based Practice Brief. National Professional Development Center on Autism Spectrum Disorders: <http://autismpdc.fpg.unc.edu/content/prompting>
Evidence-Based Practice Brief contains an overview, step-by-step directions for implementation, an implementation checklist and the evidence base for the intervention.
- Prompting. Autism Internet Module: http://www.autisminternetmodules.org/user_mod.php
In conjunction with the NPDC on ASD, this module was developed. You must sign up for a free account to access this module and the others available on this site.
- Teaching Children with Autism: <http://www.polyxo.com/discretetrial/prompting.html>
This informative article provides a description of different types of prompts and how to use them.
- What Is Prompting? <http://www.autismtreatment.info/What+Is+Prompting.aspx>
This article answers the basic question in a clear, well-written manner.

GENERAL RESOURCES

- Autism Internet Modules (AIM) www.autisminternetmodules.org
The Autism Internet Modules were developed with one aim in mind: to make comprehensive, up-to-date, and usable information on autism accessible and applicable to educators, other professionals, and families who support individuals with autism spectrum disorders (ASD). Written by experts from across the U.S., all online modules are free, and are designed to promote understanding of, respect for, and equality of persons with ASD.
Current modules are:
 - Assessment for Identification
 - Home Base
 - Peer-Mediated Instruction and Intervention (PMII)
 - Picture Exchange Communication System (PECS)
 - Pivotal Response Training (PRT)
 - Preparing Individuals for Employment
 - Reinforcement
 - Restricted Patterns of Behavior, Interests, and Activities
 - Self-Management
 - Social Supports for Transition-Aged Individuals
 - Structured Teaching
 - Structured Work Systems and Activity Organization
 - Supporting Successful Completion of Homework
 - The Incredible 5-Point Scale
 - Time Delay
 - Transitioning Between Activities
 - Visual Supports
- Interactive Collaborative Autism Network (iCAN) <http://www.autismnetwork.org>
iCAN offers free online instructional modules on autism spectrum disorder (ASD). Modules have been developed in these areas:
 - Characteristics
 - Assessment
 - Academic Interventions
 - Behavioral Interventions
 - Communication Interventions
 - Environmental Interventions
 - Social Interventions
- Indiana Resource Center for Autism (IRCA) <http://www.iidc.indiana.edu/irca/fmain1.html>
The Indiana Resource Center for Autism staff's efforts are focused on providing communities, organizations, agencies, and families with the knowledge and skills to support children and adults in typical early intervention, school, community, work, and home settings.
 - IRCA Articles <http://www.iidc.indiana.edu/irca/ftrainpapers.html>
 - IRCA Modules <http://www.iidc.indiana.edu/irca/fmodules.html>

- Texas Statewide Leadership for Autism www.txautism.net
The Texas Statewide Leadership for Autism in conjunction with the network of Texas Education Service center with a grant from the Texas Education Agency has developed a series of free online courses in autism. Please check the training page, www.txautism.net/training.html, for update lists of courses, course numbers and registration information. Current courses include the following:
 - Autism 101: Top Ten Pieces to the Puzzle
 - Autismo 101: Las 10 piezas principales del rompecabezas
 - Asperger Syndrome 101 Online
 - Asperger Syndrome 101 Online
 - Navigating the Social Maze: Supports & Interventions for Individuals with Autism Spectrum Disorders
 - Communication: The Power of Communication for Individuals with Autism Spectrum Disorders
 - Communication: The Power of Communication for Individuals with Autism Spectrum Disorders



Response Prompting as an ABA-Based Instructional Approach for Teaching Students With Disabilities

Belva C. Collins, Ya-yu Lo, Gwitaek Park,
and Kathryn Haughney

Ms. Creasy, a special education teacher, wants to teach listening comprehension to high school students with severe disabilities who use alternative or augmentative communication (AAC) devices. She hopes that her instruction will help students participate in a large-group lesson focused on reading and comprehending literature and wonders about the most effective instructional strategy she can use to accomplish this goal. Likewise, Mr. Jin is a special education teacher who has peer tutors assigned to work with high school students with moderate intellectual disability. The peer tutors attend a physical education class with the students, and Mr. Jin wonders if there is a simple but effective strategy that he can teach them to use during the class that will help the students learn targeted content.

It would be wonderful if all instruction could be errorless. Students who seldom make errors receive more positive feedback from instructors, such as “You did a great job! I am so proud of you!” rather than “I’m sorry, but that is not the correct answer.” Further, praise and other reinforcers increase the likelihood that a student will make the correct response in the future (Cooper, Heron, & Heward, 2007). On the other hand, students who make more errors are likely to practice their errors, allowing them to become ingrained (Collins, 2012). Using systematic instruction based

important to first understand its basic components. All response-prompting procedures are based on conducting instructional trials, which consist of an attentional cue, an instructional stimulus (e.g., task direction), a response, and a consequence. An *attentional cue* can be general (e.g., “Let’s begin instruction”) or specific (e.g., “If you are ready to learn, put your hand in the air”) and draws the student’s attention to the instructor or the materials. This requires an attentional response that can be general (e.g., eye contact) or specific (e.g., hand in the air) and indicates the student’s attention has been secured. Each instructional trial then consists of presenting an *instructional stimulus* (e.g., task direction) that evokes a specific response. An instructional stimulus may be verbal, signed, written, or pictorial. To ensure that a student performs a correct response during response-prompting instruction, the instructor inserts a *controlling prompt* between the stimulus and the student’s response. This consists of the amount of assistance that a student needs to perform a correct response and to receive a positive consequence. Controlling prompts are individualized for each student and each specific type of task. For example, students who cannot follow a verbal direction (e.g., “Enter the number 5 in the calculator”) may require a model prompt (e.g., demonstrate entering the numeral 5 in

Prompts are faded from instruction when the student has learned the correct response and no longer needs assistance. Finally, each student response should be followed by an appropriate *consequence* (e.g., reinforcers, such as praise, fist bump, or smiley face, for correct responses; corrective feedback, such as stating the correct verbal response or modeling the correct step of a task, for incorrect responses) that will increase the likelihood that a correct response will occur in the future. In Table 1, examples of each step of response-prompting instruction are presented.

Types of Response-Prompting Procedures

Researchers have identified six specific and distinct response-prompting procedures for teaching both academic and functional skills: (a) graduated guidance, (b) most-to-least prompting, (c) system of least prompts, (d) progressive time delay, (e) constant time delay, and (f) simultaneous prompting. As illustrated in Table 2, there is a large supporting research base to show that each procedure works for teaching various content, from single-step discrete skills to multistep chained tasks. The procedures have been effective with students with various types of abilities (e.g., no disability, intellectual disability, learning disability, emotional and behavioral disorders, autism spectrum disorder) from preschool through adulthood. The procedures also have been effective when used across various settings (e.g., school, community, home) by various instructors (e.g., special or general educators, paraprofessionals, peers without disabilities, family members, related service personnel).

The *graduated guidance* procedure typically uses physical guidance to prompt a correct response. The instructor makes a moment-by-moment decision on how much guidance a student needs to perform a correct response, shadowing the student’s movements and providing assistance only as needed, or the instructor can

Response-prompting procedures are a specific type of systematic instruction that reflects the principles of applied behavior analysis (ABA; Collins, 2012).

on the provision of prompts is one way to increase correct responses while decreasing incorrect responses. Response-prompting procedures are a specific type of systematic instruction that reflects the principles of applied behavior analysis (ABA; Collins, 2012).

Response-Prompting Instruction

In order to provide effective response-prompting instruction, it is

the calculator), and students who cannot imitate a model prompt may require a physical prompt (e.g., physically assist student in entering the numeral 5 in the calculator). Prompting can facilitate appropriate behavior by increasing the pace of instruction, which is more likely to maintain a student’s attention and reduce frustration when a student does not know how to respond (Collins, 2012; Wolery & Schuster, 1997).

Table 1. Overview of Response Prompting Procedures

Procedure (Supporting research)	Description	Implementation steps	Data collection and graphing
Graduated guidance (Wolery, Ault, & Doyle, 1992)	Uses various levels of physical guidance as needed on moment-to-moment basis	1. Secure learner's attention and deliver task direction 2. Shadow learner's movements 3. If learner fails to respond or begins to make an error, immediately use amount of physical guidance needed to make correct response 4. Praise prompted as well as unprompted responses 5. Continue until learner can perform correct response independently	• Record independent and prompted responses • Graph number or percentage of independent correct responses
Most-to-least prompting (Demchak, 1989)	Uses most-to-least intrusive prompt hierarchy across sessions (e.g., physical, model, verbal, independent)	1. Secure learner's attention and deliver task direction 2. Immediately use most intrusive prompt necessary for learner to perform correct response, praising all correct responses 3. After several sessions, move to next-less-intrusive prompt level in hierarchy 4. Repeat Step 3 5. Continue until learner can perform correct response independently across several sessions	• Record correct responses at each prompt level • Graph number or percentage of correct responses at each prompt level
System of least prompts (Doyle, Wolery, Ault, & Gast, 1988)	Uses least-to-most intrusive prompt hierarchy within sessions (e.g., independent, verbal, model, physical)	1. Secure learner's attention and deliver task direction 2. Wait a preset number of seconds (e.g., 3 seconds) for learner to respond independently 3. If learner responds correctly, give praise; if no response or error, provide least intrusive prompt in the hierarchy (e.g., verbal) and wait a set number of seconds for response 4. If no response or error, repeat Step 3 5. Continue until learner can perform correct response independently across several sessions	• Record prompt level needed for correct response • Graph number or percentage of correct independent responses (may also graph number or percentage of prompted responses)
Time delay (progressive or constant) (Walker, 2008)	Uses single controlling prompt delivered after a preset interval that increases progressively across sessions (progressive time delay) or that remains at constant interval across sessions (constant time delay)	1. Secure learner's attention and deliver task direction 2. Immediately prompt student in correct response (0-second delay interval) during first session 3. Deliver controlling prompt 4. Wait predetermined response interval (e.g., 3 seconds) 5. Praise correct responses or repeat prompt for incorrect responses or failures to respond	• Record five types of responses: correct before prompt, correct after prompt, incorrect before prompt, incorrect after prompt, or no response after prompt

(continued)

Table 1. (continued)

Procedure (Supporting research)	Description	Implementation steps	Data collection and graphing
		6. In all subsequent sessions with progressive time delay, follow Steps 1 to 5, but progressively increase number of seconds across sessions before delivering controlling prompt (e.g., 1 second during second session, 2 seconds during third session, 3 seconds during all subsequent sessions) or 7. In all subsequent sessions with constant time delay, follow Steps 1 to 5, but deliver controlling prompt at constant interval across sessions (e.g., 3 seconds)	Graph number or percentage of correct responses before prompt (may also graph number or percentage of correct responses after prompt)
Simultaneous prompting (Morse & Schuster, 2004)	Uses daily probe trials to assess student learning, followed by daily training trials using single controlling prompt delivered after a 0-second delay interval	1. First, conduct daily probe trials as follows - Secure learner's attention and deliver task direction - Wait predetermined interval (e.g., 3 seconds) for learner to respond - Do not prompt responses or correct errors 2. Following daily probe trials, conduct daily training trials as follows - Secure learner's attention and deliver task direction - Immediately deliver controlling prompt (0-second delay interval) - Praise all correct responses or correct all errors 3. Continue daily training trials until criterion is met during daily probe trials	- Record correct responses during probe trials only - Graph daily probe trial data on correct responses only

systematically fade the amount of physical guidance over time. For example, the instructor might begin by providing hand-over-hand assistance for a student first learning to use a pencil to write letters of the alphabet and fade to providing just a slight nudge to the wrist to get the student moving the pencil in the correct direction.

The *most-to-least prompting* and *system of least prompts* procedures are more systematic than graduated guidance and rely on a hierarchy of at least three prompts that will be used during instruction. In most-to-least prompting, the instructor begins by providing the most amount of assistance during initial instructional session and then systematically decreases the

amount of assistance across sessions (e.g., full physical guidance, physical model, verbal direction) until a criterion of correct independent responses (e.g., 100% unprompted responses) is reached. With this procedure, a student may take several weeks to move through the hierarchy from most assistance to independence. On the other hand, in the system of least prompts, the instructor begins each instructional trial by allowing the student the opportunity to independently perform a correct response before systematically providing an increasing amount of assistance (e.g., verbal direction, physical model, physical guidance) until the student can perform the correct response independently. The system of least

prompts is especially suited to chained tasks (e.g., multistep tasks, such as solving a complex equation or preparing a recipe) as it allows the student to complete known steps independently but wait for the amount of assistance needed on remaining steps.

Instead of providing a hierarchy of prompts, the *time delay* and *simultaneous prompting* procedures rely on a single controlling prompt—one that will result in a correct response—and this may differ across students or tasks. During the initial session of instruction of the time delay procedures, the instructor assumes that the student needs assistance and immediately provides the controlling prompt following the stimulus or task direction. For example, the instructor would show

Table 2. Recent Examples of Response Prompting From the Research

Reference	Procedures	Skills	Students/Instructor	Setting	Format
Britton, Collins, Ault, & Bausch (2017)	Constant time delay with verbal, model, or physical guidance prompts	Discrete (sight words) Chained (making drink and alphabetizing)	Secondary student with ID/paraprofessional, peer tutor	Resource room	1:1, massed trials
Collins, Evans, Galloway, Karl, & Miller (2007)	Simultaneous prompting with verbal prompt	Discrete (sight words)	Elementary, middle, and secondary students with ID/special education teachers, paraprofessional, peer tutor	General education and resource rooms	1:1, massed and distributed trials
Collins, Hager, & Galloway (2011)	Constant time delay with verbal, gesture, or model prompts	Discrete (language arts and science, core content words and concepts) Chained (math equations)	Middle school students with ID/special education teacher, paraprofessional	Resource room	1:1, massed trials
Collins, Terrell, & Test (2017)	Simultaneous prompting with verbal prompt	Chained (plant care)	Secondary student with ID/doctoral student	Greenhouse	1:1, massed trials
Creech-Galloway, Collins, Knight, & Bausch (2014)	Simultaneous prompting with verbal and model or verbal and physical guidance prompts	Chained (Pythagorean theorem problems)	Secondary students with ID/special education teacher	Resource room	1:1, massed trials
Falkenstine, Collins, Schuster, & Kleinert (2009)	Constant time delay with verbal prompts	Discrete (telling time, sight words, state abbreviations)	Secondary student with ID/special education teacher	Resource room	Small group, spaced trials
Fetko, Collins, Hager, & Spriggs (2013)	Simultaneous prompting with verbal and gestural prompts	Chained (playing UNO)	Middle school students with ID, ASD, or EBD/peer tutors without disabilities, special education teacher	Resource room	1:1, massed trials
Godsey, Schuster, Lingo, Collins, & Kleinert (2008)	Constant time delay with verbal prompts	Chained (food preparation)	Secondary students with ID/peer tutors without disabilities	Resource room	1:1, single trial
Heinrich, Collins, Knight, & Spriggs (2016)	Simultaneous prompting with verbal prompts	Discrete (math shapes, computer and biology sight words) Chained (math equations, Internet search, Punnett square)	Secondary students with ID/special education teacher (daily probe), peer tutor or paraprofessional (daily training)	General education classes (algebra, computer, biology)	1:1, distributed trials

(continued)

Table 2. (continued)

Reference	Procedures	Skills	Students/Instructor	Setting	Format
Hudson, Hinkson-Lee, & Collins (2013)	Simultaneous prompting with verbal and model prompt	Chained (writing paragraph)	Elementary students with ADHD or EBD/doctoral student	Resource room	1:1, single trial
Karl, Collins, Hager, Schuster, & Ault (2013)	Simultaneous prompting with verbal and model prompt	Discrete (reading comprehension and definitions, physics concepts) Chained (math problems and equations)	Students with ID/special education teacher	Resource room	Small group, distributed trials
Lane, Gast, Ledford, & Shepley (2017)	Progressive time delay with verbal and visual prompt or with physical prompt	Discrete (peer preferences, sharing)	Preschool students with developmental delays/doctoral student	Special education classroom	1:1 or 1:2
Manley, Collins, Stenhoff, & Kleinert (2008)	System of least prompts with hierarchy (verbal, verbal and model, verbal and physical guidance)	Chained (making a telephone call)	Elementary students with ID/special education teacher	Resource room	1:1, single trials
Pennington, Collins, Stenhoff, Turner, & Gunselman (2014)	Simultaneous prompting with gestural prompt	Chained (sentence writing using a computer program)	Elementary students with ASD/special education teacher, paraprofessional	Resource or therapy room	1:1, massed trials
Riggs, Collins, Kleinert, & Knight (2013)	Constant time delay with verbal and model prompt	Discrete (heredity traits) Chained (Punnett square)	Secondary students with ID/special education teacher	Resource room	1:1 and small group, massed trials
Tekin-Iftar, Collins, Spooner, & Olcay-Gul (2017)	Simultaneous prompting with verbal prompt	Discrete (health core content facts)	Middle school students with ASD/general education health teachers	General education health class	1:1, distributed trials

Note. ID = intellectual disability; ASD = autism spectrum disorder; EBD = emotional and behavioral disorders; ADHD = attention deficit hyperactivity disorder.

a flashcard of a sight word, ask the student to read the word, and immediately state the word as a model for the student's response. This is called a 0-second delay trial because no time elapses between the stimulus and the prompt. Once the instructor has conducted a preset number of 0-second delay sessions (e.g., one session of five trials each), the instructor begins to increase the delay interval by allowing an opportunity for the student to perform a correct independent response before providing the controlling prompt.

With the *progressive time delay* procedure, the instructor provides a set number of 1-second delay sessions for the student to respond before providing the controlling prompt, then increases the delay interval to 2 seconds for a set number of sessions. This format of increasing the delay interval by 1 second at a time across sessions continues until the student reaches a predetermined delay interval (e.g., 5 seconds). With the *constant time delay* procedure, the instructor immediately follows 0-second delay sessions by

providing a constant delay interval (e.g., 5 seconds) for the student to respond before providing the controlling prompt across all subsequent sessions. In both time delay procedures, correct responses before or after the prompt are always praised, incorrect responses before the prompt are followed by a reminder to wait for a prompt if the student is unsure of the correct response, and incorrect responses or failures to respond after the prompt are followed by corrective feedback (e.g., "No, the

correct answer is rectangle”). The controlling prompt is naturally faded when the student reaches criterion of a set number of sessions of correct responses before the prompt. Whereas the progressive time delay procedure may be most appropriate for students who do not know how to wait for a prompt, the constant time delay procedure may be easier for instructors to implement as it is not necessary to remember which delay interval is being used during each session.

With *simultaneous prompting*, the instructor always begins by conducting probe (i.e., assessment or test) trials to determine if the student has mastered the content to be taught. If the daily probe session results reveal that the student has not mastered the content, the instructor then conducts daily 0-second delay training (i.e., instructional) trials in the same manner as the initial trials of the time delay procedures, by immediately delivering the controlling prompt following the stimulus or task direction and without allowing the opportunity for the student to respond independently. The sequence of daily probe trials followed by daily training trials continues until the student meets criterion (e.g., three sessions of 100% correct independent responses) during daily probe trials.

Implementing Response Prompting Procedures Efficiently

In addition to being effective, response-prompting strategies can be an efficient way to teach. Response-prompting strategies often require less instructional time to implement than other procedures because instructors do not have to spend a lot of time correcting errors. There may be reasons to use a hierarchy of prompts (e.g., physical assistance, modeling, direct verbal, indirect verbal) to teach some skills to some students. For example, some students may know how to make a correct response on specific steps of a chained task but may need various levels of assistance on other steps; an instructor using the system of least prompts provides prompts only as needed. Response-prompting

procedures that use a single controlling prompt, however, are typically more efficient because they take less time to implement (Ault, Wolery, Gast, Doyle, & Eizenstat, 1988).

Another way to make response-prompting procedures more efficient is to add content not directly targeted for instruction (i.e., nontargeted information) to instructional trials and to use strategies that facilitate generalization. Teachers may increase the amount of content that a student learns during systematic instruction by adding nontargeted information as *instructive feedback* during instructional trials (Werts, Wolery, Holcombe, & Gast, 1995). For example, the instructor can state related nontargeted information when praising a student’s response, such as “That is correct—this shape is a square. *S-Q-U-A-R-E*. A square has four equal sides and four 90-degree angles.” In addition to learning to recognize a square, the student may learn to spell the word and to state the characteristics of a square after repeated exposure to this information. Another way to facilitate the learning of nontargeted information is through *observational learning*, where each student taught in a group has the opportunity to observe the content being taught to other members of the group. For example, a student being taught to recognize circles also may learn to recognize squares by observing this being taught to another group member and may even learn instructive feedback (e.g., spelling of the word) presented to the other student as well (Campbell & Mechling, 2009).

When using a response-prompting procedure, the instructor can consider ways to increase the likelihood that a student will generalize a learned response across settings, people, and materials as first described by Stokes and Baer (1977). For example, an instructor can facilitate generalization of shape recognition (a) across *materials* by presenting multiple examples (e.g., square boxes, square windows, square picture frames, and square books) during instruction, (b) across *settings* by pointing out square objects as they appear naturally in various school settings, and (c) across *people* by

varying instructors (e.g., teachers, peers, paraprofessionals). The goal is for a student to use a new skill with any material in any environment with any person. Thus, it is important to, at least periodically, assess students for generalization across novel materials, settings, and persons while instruction is occurring as well as when students reach criterion (Cooper et al., 2007). If generalization fails to occur, additional instruction that systematically addresses generalized outcomes may be needed (e.g., teaching with two or three novel materials and then assessing with an additional novel material; teaching in a natural setting versus a simulation and then assessing in an additional novel setting).

Countering Myths of Instruction Using Response-Prompting Procedures

Despite the reasons to use response-prompting procedures, there are a number of myths teachers may believe that can decrease their willingness to teach in this way. These include the belief that response prompting can be delivered only by a trained special education teacher in a special education classroom in a massed trial format when teaching discrete skills (e.g., teaching multiple repetitions of a one-step response, such as a sight word) in a one-to-one format. In reality, response-prompting procedures are more flexible and less regimented than what teachers may believe.

There is also a myth that students taught with response-prompting procedures become dependent on prompts, but there are strategies to decrease this likelihood, such as fading prompts and providing more specific praise or individualized reinforcement (e.g., time to engage in a preferred activity) for unprompted responses than for prompted responses (i.e., differential reinforcement). The examples shown in Table 2 are but a few from the research literature over the past 10 years that have shown that response-prompting procedures can be used to teach a variety of skills to a variety of

Figure 1. Example of lesson plan using the system-of-least-prompts procedure.

Lesson Plan 1: Listening Comprehension											
Core Content Standards											
<i>Literacy, Speaking and Listening</i>											
• ELA, LITERACY-Students will read and comprehend literature.											
Behavioral Objective											
• After listening to a read aloud passage, each participant will correctly respond to 5 of 5 WH-comprehension questions by independently pointing to the correct response option on a response board for 3 out of 3 sessions.											
Instructional Context											
Instruction will occur in a group format during a regularly scheduled Social Studies lesson following an adapted informational text read aloud. Target students, Brett, Quinn, and Rhys, are each seated in groups with 3 or 4 other peers, one of whom is a peer (or a paraprofessional) trained to support their learning and AAC turn-taking.											
Instructional Materials											
Data sheet and pen; adapted social studies text; response boards with at least 4 picture-supported response options (always including at least one “near” example from the same text) and a WH question anchor chart (including the word, symbol, basic definition, and example of Who, What, When, Where, and Why).											
Instructional Procedure: System of Least Prompts (5 trials per session)											
Using the <i>system of least prompts</i> procedure, each instructional trial will proceed as follows.											
Attentional Cue	• “Ready to answer questions about the story we read?” • Gain target student’s attention with a verbal directive (e.g., “Quinn, look at me.”) and a gesture.										
Task Direction	• Individually ask a WH question relevant to the lesson (e.g., What did we read about? Who was the story about? When did the story take place? Where did the story take place?).										
Prompting Procedure	<table> <tr> <td>Correct</td><td>Independent response within 3 seconds</td></tr> <tr> <td>Prompt level 1</td><td>If student fails to give correct response within 3 seconds, communication partner will reread relevant sentence form the adapted text (verbal prompt).</td></tr> <tr> <td>Prompt level 2</td><td>If student still fails to give correct response within 3 seconds, communication partner will reference WH question chart to remind student about type of question being asked and then reread relevant sentence (verbal prompt plus explanation).</td></tr> <tr> <td>Prompt level 3</td><td>If student still fails to give correct response within 3 seconds, communication partner will model referencing the WH chart (e.g., A <i>Who</i> question...) and then reread relevant sentence (verbal prompt plus model).</td></tr> <tr> <td>Prompt level 4</td><td>If student still fails to give correct response within 3 seconds, communication partner will physically guide student in identifying correct answer on communication device (verbal prompt plus physical guidance).</td></tr> </table>	Correct	Independent response within 3 seconds	Prompt level 1	If student fails to give correct response within 3 seconds, communication partner will reread relevant sentence form the adapted text (verbal prompt).	Prompt level 2	If student still fails to give correct response within 3 seconds, communication partner will reference WH question chart to remind student about type of question being asked and then reread relevant sentence (verbal prompt plus explanation).	Prompt level 3	If student still fails to give correct response within 3 seconds, communication partner will model referencing the WH chart (e.g., A <i>Who</i> question...) and then reread relevant sentence (verbal prompt plus model).	Prompt level 4	If student still fails to give correct response within 3 seconds, communication partner will physically guide student in identifying correct answer on communication device (verbal prompt plus physical guidance).
Correct	Independent response within 3 seconds										
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Prompt level 3	If student still fails to give correct response within 3 seconds, communication partner will model referencing the WH chart (e.g., A <i>Who</i> question...) and then reread relevant sentence (verbal prompt plus model).										
Prompt level 4	If student still fails to give correct response within 3 seconds, communication partner will physically guide student in identifying correct answer on communication device (verbal prompt plus physical guidance).										
Consequence	• The teacher or communication partner will praise correct response whether it is independent or prompted (e.g., “Great job answering the question!”)										
Data Collection											
On the data sheet, the paraprofessional or peer will circle the most intrusive prompt level (P1, P2, P3, P4) needed for the student to make the correct response on questions 1-5 using the key provided in the following data sheet.											

(continued)

Figure 1. (continued)

System of Least Prompts Data Sheet					
Name	Quinn		Setting	Social Studies, Rm 203	
Peer	Rhys		Skill	Listening Comprehension	
Date	9/12/18		Time	9:30 a.m.	
Code Key	+ Independent Correct				
Prompt Levels for	P1	Reread relevant sentence from adapted text			
Multiple Choice	P2	Reference WH question anchor chart, reread relevant sentence			
	P3	Model referencing WH question anchor chart, reread answer			
	P4	Physical guidance to identify correct answer			
Circle based on response					
	Say ...	C	Prompt Level	Notes	
Question 1	Who did you read about? (wait 6 sec)	+	P1 P2 P3 P4		
Question 2	What did Simon do in the story? (wait 6 sec)	+	P1 P2 P3 P4	Vocal - AAC	
Question 3	Where does Simon live? (wait 6 sec)	+	P1 P2 P3 P4		
Question 4	What does Simon always use? (wait 6 sec)	+	P1 P2 P3 P4		
Question 5	Who did Simon meet? (wait 6 sec)	+	P1 P2 P3 P4	"Simon"	
No. Ind. Correct	2 / 5				
Mastery (5 / 5)	No Yes				

Maintenance

Once criterion has been met for one day, praise for correct responses will be delivered intermittently (e.g., after 2 or 3 correct responses instead of after every correct response) for two additional days to facilitate maintenance. Maintenance of listening comprehension will be monitored at 1-week intervals for 1 month.

Generalization

All students will be exposed to multiple exemplars of adapted texts during social studies lessons to facilitate generalization of listening comprehension to new content. Brett will be assessed on listening comprehension demonstrated in social studies during lessons in other inclusive classes.

Behavior Management

All students will be intermittently praised for remaining on task; Rhys will receive regular praise for staying in his seat or his standing area behind his desk while answering questions.

Lesson Variations and Extension

As a variation, listening comprehension instruction can be implemented across various subjects from science and literacy to job training. As an extension, a preference question (e.g., "Why did you like this topic? Which topic did you like best?") may be added to determine student preference and connect to their personal experiences. This preference question can be added to personal AAC supports to aid in class discussion.

students across ages and disabilities by a variety of instructors in a variety of settings.

After doing some research, Ms. Creasy and Mr. Jin decide to use instructional strategies based on the principles of ABA. Ms. Creasy chooses a system-of-least-prompts procedure, whereas Mr. Jin chooses a simultaneous-prompting procedure.

Ms. Creasy works with her paraprofessional to teach listening comprehension to a small group of students with severe disabilities who use augmentative communication devices (Figure 1). Mr. Jin develops a lesson plan (Figure 2) that includes the presentation of nontargeted information (e.g., benefits of aerobic exercise) as instructive feedback during instructional trials.

Guidelines for Implementing Response-Prompting Procedures

Almost anyone can learn to teach using a response-prompting procedure. It is important, however, to follow four key guidelines that have been suggested by researchers who are experienced in using response-prompting procedures (Collins, 2012; Wolery, Ault, & Doyle, 1992). First, select a response-prompting procedure

Figure 2. Example of lesson plan using simultaneous-prompting procedure.

Lesson Plan 2: Health Core Content

Core Content Standards

Health: Students will summarize the benefits and examples of regular physical activity.

Behavioral Objective

When asked the following two targeted health questions 5 times each, Barbara will respond correctly 10 out of 10 times for 3 days.

- “Why is aerobic exercise important?” (“Aerobic exercise can make your body healthy and fit.”)
- “Why is stretching important?” (“Stretching can improve the range of motion of your body.”)

Instructional Context

Daily probe trials will be conducted in a one-to-one format by the special education teacher each day in the resource room. This will be followed by daily training trials conducted by a peer tutor in a one-to-one format in the school gym while walking laps with Barbara at the beginning of each physical education class.

Instructional Materials

Data sheets and a pen for the teacher; a laminated instruction card (see below) for the peer tutor. The numbers 1-5 on the card are for the peer tutor to fold back each day to keep track of number of trials.

How to instruct your peer! (Instruction card)					
Attentional cue 		Task direction/Question 		Immediate deliver prompt (Target) 1. Aerobic exercise can make your body healthy and fit. 2. Stretching can improve the range of motion of your body	
Request repeat 		Feedback on target 		Immediate deliver prompt (Non-target) 1. Walking is a good example of aerobic exercise. 2. Show your peer like this 	
1	2	3	4	5	

Instructional Procedures: Simultaneous Prompting (5 trials per session)

The special education teacher will first conduct each *daily probe trial* in the resource room as follows:

Attentional Cue

- “Are you ready? Look at me” or state Barbara’s name to get her attention.

Task Direction on Targeted Content

- Alternate asking each of the targeted core content questions five time each.

Consequence

- Provide verbal praise for correct responses within 3 seconds (e.g., “Great job responding with the answer”); ignore incorrect responses or failures to respond within 3 seconds.

Task Direction on Nontargeted Information

- Alternate asking nontargeted core content questions five time each (See questions in training trials).

Consequence

- Provide verbal praise for correct responses (e.g., “Great job responding with the answer”) within 3 seconds; ignore incorrect responses or failures to respond within 3 seconds.

The peer tutor will conduct each *daily training trial* during physical education class as follows:

Attentional Cue

- “Are you ready? Look at me.” or state Barbara’s name to get her attention.

Task Direction 1

- Ask first question, “Why is aerobic exercise important?”

Verbal Prompt

- Immediately state correct answer (i.e., “Aerobic exercise can make your body healthy and fit.”) and ask Barbara to repeat answer.

Consequence

- Deliver verbal praise (e.g., “Great job repeating the answer”) or error correction (e.g., “No, the correct answer is ...”).

Instructive Feedback

- State nontargeted information (i.e., “Walking is a good example of aerobic exercise.”).

(continued)

Figure 2. (continued)

Task Direction 2

Verbal Prompt

Consequence

Instructive Feedback

Remaining Trials

Data Collection

- Ask second question, “Why is stretching important?”
- Immediately state correct answer (i.e., “Stretching can improve the range of motion of your body.”) and ask Barbara to repeat answer.
- Deliver verbal praise (e.g., “Great job repeating the answer”) or error correction (e.g., “No, the correct answer is ...”).
- Model nontargeted information (i.e., Show Barbara upper body stretching postures).
- Repeat procedure for 5 trials, alternating between questions.

The special education teacher will collect data using the data sheet below to assess learning during *daily probe trials*. On the data sheet, the teacher will circle a plus sign for correct responses, a minus sign for incorrect responses, and a zero for no responses. (NOTE: Only the data on targeted information counts toward criterion. If a student does not reach criterion on nontargeted information, this will be targeted for direct instruction in the future.)

SIMULTANEOUS PROMPTING DATA SHEET					
Name: <i>Barbara</i>	Core content: <i>Health</i>	Date: <i>October 24</i>			
Instructor: <i>Mr. Park</i>	Setting: <i>Resource Room</i>	Time: <i>9:00 a.m.</i>			
Circle the student response for each trial					
<i>Targeted</i> core content	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
Q: Why is aerobic exercise important? A: Aerobic exercise can make your body healthy and fit.	⊕ - 0	+ - ①	⊕ - 0	⊕ - 0	+ - ①
Q: Why is stretching important? A: Stretching can improve the range of motion of your body.	⊕ - 0	⊕ - 0	+ - ①	⊕ - 0	⊕ - 0
NUMBER CORRECT (+):	7/10				
<i>Nontargeted</i> core content	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
Q: What is an example of aerobic exercise? A: Walking.	+ ① 0	+ ① 0	⊕ - 0	⊕ - 0	⊕ - 0
Q: What is an example of stretching? A: Upper stretching postures.	+ - 0	⊕ - 0	⊕ - 0	⊕ - 0	⊕ - 0
NUMBER CORRECT (+):	8/10				

Maintenance

Once Barbara has met the criterion of 100% correct responses on targeted core content for 1 day, the special education teacher will fade the amount of verbal praise (e.g., after every 2 or 3 correct responses) for 2 days. After 2 weeks, Barbara will be reassessed with the same questions to determine whether she has maintained the core content.

Generalization

The teacher will assess generalization by having Barbara’s peers and parents ask the same questions across settings (e.g., home, classrooms).

Behavior Management

The peer tutor will periodically praise Barbara for paying attention during daily training trials and the teacher will periodically praise Barbara for paying attention during daily probe trials.

Lesson Variations and Extension

Once Barbara has mastered the targeted content, the special education teacher will collaborate with peers without a disability to teach Barbara other additional core content (such as shooting a basketball, catching a baseball, or hitting a volleyball) across other physical activities. If nontargeted information has not been mastered, this can be taught more directly.

that is the best fit for the student and the task. The procedures that initially use more intrusive prompts (i.e., those that provide a great amount of assistance) and provide less opportunity for independence (e.g., graduated guidance, most-to-least prompting) may be better suited to young children or students with more significant disabilities as students

Third, let data guide instruction. It is important that instructors who use response-prompting procedures monitor formative data closely, consistently, and frequently. Data need to be recorded on data collection sheets as instruction occurs and then graphed immediately following daily instruction. This allows instructors to use visual analysis to

There is a large amount of research to support the effectiveness of response-prompting procedures, and the research base continues to grow.

are not required to wait for a prompt before responding, be imitative, or know how to follow directions. Likewise, the procedures that use a single controlling prompt (e.g., time delay, simultaneous prompting) may be faster paced and result in less frustration for students with short attention spans as the instructor does not have to implement a hierarchy of prompts until a student learns a skill. In addition, procedures like time delay or simultaneous prompting may be better suited for teaching discrete skills, like math facts or sight words, whereas the system of least prompts may be better suited for teaching chained tasks where the student can complete known steps but wait for assistance on unknown steps.

Second, if two procedures are equally effective, select the one that is most parsimonious or efficient. Applying the principle of parsimony (Whaley & Surratt, 1968), instructors should pay attention to what works with each student and which procedure is easiest to implement. For example, if the time delay and the system-of-least-prompts procedures are equally effective in teaching a particular task to a particular student, then the teacher might select the time delay procedure because it requires using only a single controlling prompt instead of keeping up with an entire hierarchy of prompts. The efficiency of a procedure can be determined by considering which procedure results in the desired outcome in the smallest number of instructional sessions or with the lowest rate of errors.

quickly determine if instruction is effective. If there is little to no progress over several sessions, the instructor can “tweak” instruction by using a different reinforcer that is more motivating to the student, change to a more intrusive prompt that is more likely to result in a correct response, or consider whether the student needs to stop and learn prerequisite skills or be taught a complex skill in more detailed steps or smaller chunks of steps (Collins, 2012).

Fourth, increase efficiency of instruction when possible. As previously noted, efficiency can be increased by adding nontargeted information as instructive feedback during instructional trials, teaching in a small-group format to facilitate observational learning, and using strategies that facilitate generalization. When a student reaches criterion on a task, the instructor should do a quick assessment to see if the student has learned additional, nontargeted information. If this has occurred, future instructional time can be devoted to teaching other skills. If the student has not learned the nontargeted information, it is possible that future instructional time on this content may not require much additional instruction (Wolery, Schuster, & Collins, 2000). The instructor also should assess generalization to determine if the student can perform a learned task across different (and novel) materials, settings, or people. If the student fails to generalize, additional instruction across these variables is needed.

Summary

Response-prompting procedures are a form of systematic direct instruction based on the principles of ABA. There is a large amount of research to support the effectiveness of response-prompting procedures, and the research base continues to grow. Although earlier professional literature on response-prompting procedures focused more on functional skills taught in special education settings, more current research has included core content taught in inclusive settings. Response-prompting procedures are simple enough for almost anyone to learn and are flexible enough to be used in a variety of settings with a variety of skills.

References

- Ault, M. J., Wolery, M., Gast, D. L., Doyle, P. M., & Eizenstat, V. (1988). Comparison of response prompting procedures in teaching numeral identification to autistic subjects. *Journal of Autism and Developmental Disorders*, 18, 627–636. doi:10.1007/BF02211880
- Britton, N. S., Collins, B. C., Ault, M. J., & Bausch, M. E. (2017). Using a constant time delay procedure to teach support personnel to use a simultaneous prompting procedure. *Focus on Autism and Other Developmental Disabilities*, 32, 102–113. doi:10.1177/1088357615587505
- Campbell, M. L., & Mechling, L. C. (2009). Small group computer-assisted instruction with SMART board technology: An investigation of observational and incidental learning of nontarget information. *Remedial and Special Education*, 30, 47–57. doi:10.1177/0741932508315048
- Collins, B. C. (2012). *Systematic instruction for students with moderate and severe disabilities*. Baltimore, MD: Brookes.
- Collins, B. C., Evans, A., Galloway, C. G., Karl, A., & Miller, A. (2007). A comparison of the acquisition and maintenance of teaching functional and core content in special and general education settings. *Focus on Autism and Other Developmental Disabilities*, 22, 220–233.
- Collins, B. C., Hager, K. D., & Galloway, C. C. (2011). The addition of functional content during core content instruction with students with moderate disabilities. *Education and Training in Autism and Developmental Disabilities*, 46, 22–39.

- Collins, B. C., Terrell, M., & Test, D. W. (2017). Using a simultaneous prompting procedure to embed core content when teaching a potential employment skill. *Career Development and Transition for Exceptional Individuals*, 40, 36–44. doi:10.1177/2165143416680347
- Cooper, J. O., Heron, T. E., & Heward, W. L. (2007). *Applied behavior analysis* (2nd ed.). Upper Saddle River, NJ: Pearson/Merrill-Prentice Hall.
- Creech-Galloway, C., Collins, B. C., Knight, V., & Bausch, M. E. (2014). Using simultaneous prompting with an iPad to teach the Pythagorean theorem to adolescents with moderate and severe disabilities. *Research and Practice in Severe Disabilities*, 38, 222–232. doi:10.1177/154079691303800402
- Demchak, M. (1989). A comparison of graduated guidance and increasing assistance in teaching adults with severe handicaps leisure skills. *Education and Training in Developmental Disabilities*, 24, 45–55.
- Doyle, P. M., Wolery, M., Ault, M. J., & Gast, D. L. (1988). System of least prompts: A systematic review of procedural parameters. *Journal of the Association for Persons with Severe Handicaps*, 13, 28–40. doi:10.1177/154079698801300104
- Falkenstine, K. J., Collins, B. C., Schuster, J. W., & Kleinert, K. (2009). Presenting chained and discrete tasks as nontargeted information when teaching discrete academic skills through small group instruction. *Education and Training in Developmental Disabilities*, 44, 127–142.
- Fetko, E. E., Collins, B. C., Hager, K. D., Schuster, J. W., & Spriggs, A. (2013). Using peer tutors and a simultaneous prompting procedure to teach a leisure skill to students with disabilities. *Education and Training in Autism and Developmental Disabilities*, 48, 400–411.
- Godsey, J. R., Schuster, J. W., Lingo, A. S., Collins, B. C., & Kleinert, H. L. (2008). Peer-implemented time delay procedures on the acquisition of chained tasks by students with moderate and severe disabilities. *Education and Training in Developmental Disabilities*, 43, 111–122.
- Heinrich, S., Collins, B. C., Knight, V., & Spriggs, A. D. (2016). Embedded simultaneous prompting procedure to teach STEM content to high school students with moderate disabilities in an inclusive setting. *Education and Training in Autism and Developmental Disabilities*, 51, 41–54.
- Hudson, T. H., Hinkson-Lee, K., & Collins, B. C. (2013). Teaching paragraph composition to student with emotional/behavioral disorders using the simultaneous prompting procedure. *Journal of Behavioral Education*, 22, 139–156. doi:10.1007/s10864-012-9167-8
- Karl, J., Collins, B. C., Hager, K. D., Schuster, J. W., & Ault, M. J. (2013). Teaching core content embedded in a functional activity to students with moderate cognitive disabilities using a simultaneous prompting procedure. *Education and Training in Autism and Developmental Disabilities*, 48, 363–378.
- Lane, J. D., Gast, D. L., Ledford, J. R., & Shepley, C. (2017). Increasing social behaviors in young children with social-communication delays in a group arrangement in preschool. *Education & Treatment of Children*, 40, 115–144. doi:10.1353/etc.2017.0007
- Manley, K., Collins, B. C., Stenhoff, D. M., & Kleinert, H. (2008). Using a system of least prompts procedure to teach telephone skills to elementary students with cognitive disabilities. *Journal of Behavioral Education*, 17, 221–236. doi:10.1007/s10864-008-9065-2
- Morse, T. E., & Schuster, J. W. (2004). Simultaneous prompting: A review of the literature. *Education and Training in Developmental Disabilities*, 39, 153–168.
- Pennington, R., Collins, B. C., Stenhoff, D. M., Turner, K., & Gunselman, K. (2014). Using simultaneous prompting to teach generative writing to students with autism. *Education and Training in Autism and Developmental Disabilities*, 49, 396–414.
- Riggs, L., Collins, B. C., Kleinert, H., & Knight, V. (2013). Teaching principles of heredity to high school students with moderate and severe disabilities. *Research and Practice in Severe Disabilities*, 38, 30–43. doi:10.2511/027494813807046971
- Stokes, T. F., & Baer, D. M. (1977). An implicit technology of generalization. *Journal of Applied Behavior Analysis*, 10, 349–367.
- Tekin-Iftar, E., Collins, B. C., Spooner, F., & Olcay-Gul, S. (2017). Teaching teachers to use systematic instruction to teach core content to students with ASD. *Teacher Education and Special Education*, 42, 225–245. doi:10.1177/0888406417703751
- Walker, G. (2008). Constant and progressive time delay procedures for teaching children with autism: A literature review. *Journal of Autism and Developmental Disorders*, 38, 261–275. doi:10.1007/s10803-007-0390-4
- Werts, M. G., Wolery, M., Holcombe, A., & Gast, D. L. (1995). Instructive feedback: Review of parameters and effects. *Journal of Behavioral Education*, 5, 55–75. doi:10.1007/BF02110214
- Whaley, D. L., & Surratt, S. L. (1968). *Attitudes of science*. Kalamazoo, MI: Behaviordelia.
- Wolery, M., Ault, M. J., & Doyle, P. M. (1992). *Teaching students with moderate to severe disabilities: Use of response prompting strategies*. New York, NY: Longman.
- Wolery, M., & Schuster, J. W. (1997). Instructional methods for students who have significant disabilities. *The Journal of Special Education*, 37, 61–79. doi:10.1177/002246699703100106
- Wolery, T. D., Schuster, J. W., & Collins, B. C. (2000). Effects of future learning of presenting non-target stimuli in antecedent and consequent conditions. *Journal of Behavioral Education*, 10, 77–94. doi:10.1023/A:1016679928480

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